

SACS



Collapse

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The SACS module Collapse is a large deflection, elasto-plastic, nonlinear finite element system for structures. The program is fully integrated into the SACS suite of programs and uses the same input data as that for a standard SACS IV/PSI analysis. No new modeling is required to conduct a full plastic collapse analysis of a structure.

1.2 PROGRAM FEATURES

The Collapse program requires no special modeling and only minimal additional input specified in a Collapse input file. Some of the main capabilities and features of the program are as follows:

- Linear and nonlinear material behavior.
- Nonlinear plastic pile/soil foundation including standard T-Z and P-Y data.
- Includes member global/local buckling including 8 or more hinge points per member.
- Accounts for segmented elements automatically.
- Includes tubular joint flexibility, joint plasticity and joint failure due to excess strain.
- Includes strain hardening and residual stress.
- Material properties default to perfectly elastic/perfectly plastic.
- User defined nonlinear spring support elements.
- Sequential load stacking capability with user controlled load incrementation, includes both loading and unloading capabilities.
- Load cases may contain loading and/or specified displacements.
- Creates analysis results file that is read by Collapse View program which shows failure progression and the gradual plastification and collapse mechanism graphically.

1.3 PROGRAM STRUCTURE

The basic procedure used by the Collapse program to perform the nonlinear analysis is as follows:

1.3.1 Beam Elements

Beam element stiffness is developed using second order effects with nonlinear material properties. Each beam is automatically discretized by using sub-segments along the member length. Each length sub-segment is additionally divided into sub-elements through the beam cross section to define the cross section shape.

The beam element is treated as a superelement whose stiffness is defined by the stiffnesses of its sub-elements. While the intermediate nodes along the member are reduced for stiffness, the deflected shape of the element is represented by all sub-segments.

Note: Beam elements designated as elastic elements are treated as a single element.

By default, non-segmented beam elements are divided into eight sub-segments along the length of the element while segmented beam elements are divided into sub-segments according to the change in cross section. The number of sub-elements per sub-segment is based on the element cross section type. For tubular beams for example, each sub-segment is divided into 12 sub-elements around the circumference. For other cross section shapes similar cross section representations are constructed.

For any stiffness iteration, each sub-element is checked for plasticity using a von Mises stress surface. When the stresses in a sub-element exceed the material elastic limit, the sub-element is considered plastic, thus allowing for gradual plastification of the beam cross section. When all sub-elements of a particular sub-segment become plastic, a temporary hinge is formed at that sub-segment.

For beam elements, the stress history of each sub-element is monitored for plasticity, strain hardening and unloading. The beam deflected shape is calculated at the member ends and along its length at each sub-segment. Member elastic and plastic buckling is automatically calculated using the beam deflected shape and the plasticity of the member sub-segments. Local tubular buckling is determined using the total strain in the cross section and is treated as a permanent hinge after it develops.

1.3.2 Plate Elements

Plate elements are divided into 5 sub-layers through the thickness to allow for gradual plastification. Plate elements are not divided into sub-elements along the surface length and width of the plate.

Each plate sub-layer may become plastic and plate buckling and snap through are included in the solution. Because the stress history of each sub-layer is monitored, the plate element retains plastic deformation and residual stress.

1.3.3 Tubular Connections

Tubular joint flexibility options include 1) Fessler's empirical formulas 2) Single brace formulation due to Buitrago 3) MSL formulation

Tubular connection failure criterion can be introduced via 1) Modified ultimate LRFD strength formulation as per API code of practice, 2) Joint strength check as per NORSOK code of practice, 3) MSL formulation. The brace stiffness is automatically removed from the analysis at the onset of a connection failure.

Local buckling options include 1) Marshall and Gates minimum strain criteria 2) API Bulletin 2U, 3) API LRFD 1st Edition code of practice 4) ISO 19902 code of practice. A permanent hinge is introduced along a member where the local buckling criterion is exceeded.

1.3.4 Member Distributed Loading

Member distributed loads are treated as equivalent point loads acting at the end joints of the member sub-segments. This allows for an accurate representation of distributed loading.

1.3.5 Foundations

The Collapse solution may include the effects of a nonlinear pile/soil foundation. Tubular pile elements are segmented along the length and around the circumference and are treated in the same manner as tubular members. Soil data is represented with standard T-Z and P-Y data in PSI format.

1.3.6 Solution Technique

The solution process involves three levels of iteration. For any global load increment, a beam-column solution is performed for each plastic member using the cross section sub-element details. The global stiffness iteration is then performed including the effects of connection flexibility, plasticity and failure and the foundation stiffness iteration includes the nonlinear pile/soil effects.

During any global solution iteration, the deflected shape of the structure is determined and compared to the displacements of the previous solution iteration. If convergence is not achieved, the new global displacements of the joints along with the beam internal and external loads are used to recalculate the elemental stiffness matrices. The structural stiffness iteration is then repeated including the effects of the foundation until the displacements meet the convergence tolerance.

1.3.6.1 Foundation Solution

The solution of the pile/soil foundation requires an iterative procedure. Initially, soil forces and stiffness is calculated assuming deflections and rotations are zero along the full length of the pile. For the given pilehead displacement, the pile deflections and rotations are then determined. New soil forces and stiffness is calculated based on these new displacements and rotations.

Using the segment deflections and rotations, the program computes the pile segment internal loads then calculates the pile segment plasticity. The resulting plastic forces are then applied to the pile segment for the next iteration. This procedure is repeated until all of the deflections and rotations along the pile length have converged.

At the final deflected position, the program calculates the pilehead stiffness matrix by incrementally varying the pilehead deflections and rotations and computing the pilehead restraining forces and moments. The resulting pilehead plastic forces are transformed into the global coordinates and added to the global plastic force vector for the next global increment or iteration.

1.3.7 Analysis Considerations

The Collapse module is capable of handling most structural problems where plasticity may occur through large deflections. Some obvious applications include Progressive Collapse Analysis, Ship Impact, Dropped Object Studies and general Safety Case Studies. Some basic considerations in conducting such analysis are outlined below:

1.3.7.1 Progressive Collapse Analysis

The 'Plastic Collapse' mode of assessment offers an improved design concept over linear 'Elastic' theory for the analysis/re-analysis of structures. The basic concept of the Plastic Collapse Analysis is as follows:

The load is applied to the structure incrementally. The nodal displacements and element forces are calculated for each load step and the stiffness matrix is updated. When the stress in a member reaches

the yield stress plasticity is introduced. The introduction of plasticity reduces the stiffness of the structure and additional loads due to subsequent load increments will be redistributed to adjacent members to the members that have gone plastic. This phenomenon (progressive collapse of members) will continue until the structure as a whole will collapse or is 'Pushed Over'.

For large offshore structures the analysis can be highly CPU intensive since each element is subdivided into eight sub segments and for tubular elements each sub-segment is further divided into 12 sub-elements around the circumference. Collapse run time can be decreased by modeling parts of the structure which have little or no contribution to the overall stiffness of the structure (such as boat landings for example) as dummy structures. All elements contained in a dummy structure are removed by the Seastate module and the loads on the dummy structure are transferred to the main structure before the Collapse analysis is initiated.

Elements whose stiffness may be of significance to the overall behavior of the structure but which are not structurally important (such as conductors and conductor guides, wishbone elements, topsides elements ...etc.) should be kept elastic throughout the loading history.

Further cut backs in run time can be achieved by pre-combining loads wherever possible to cut down the number of loads in a load sequence. Also, a structure undergoing a high level of nonlinear behavior can result in an increasing number of iterations for the solution to converge. In such cases it is better to reduce the step size than to increase the maximum iteration limit. Reducing the step size effectively linearizes the problem and results in decrease in the number of iterations and therefore a decrease in runtime.

1.3.7.2 Ship Impact

A ship impact scenario involves transference of ships kinetic energy into strain energy resulting from:

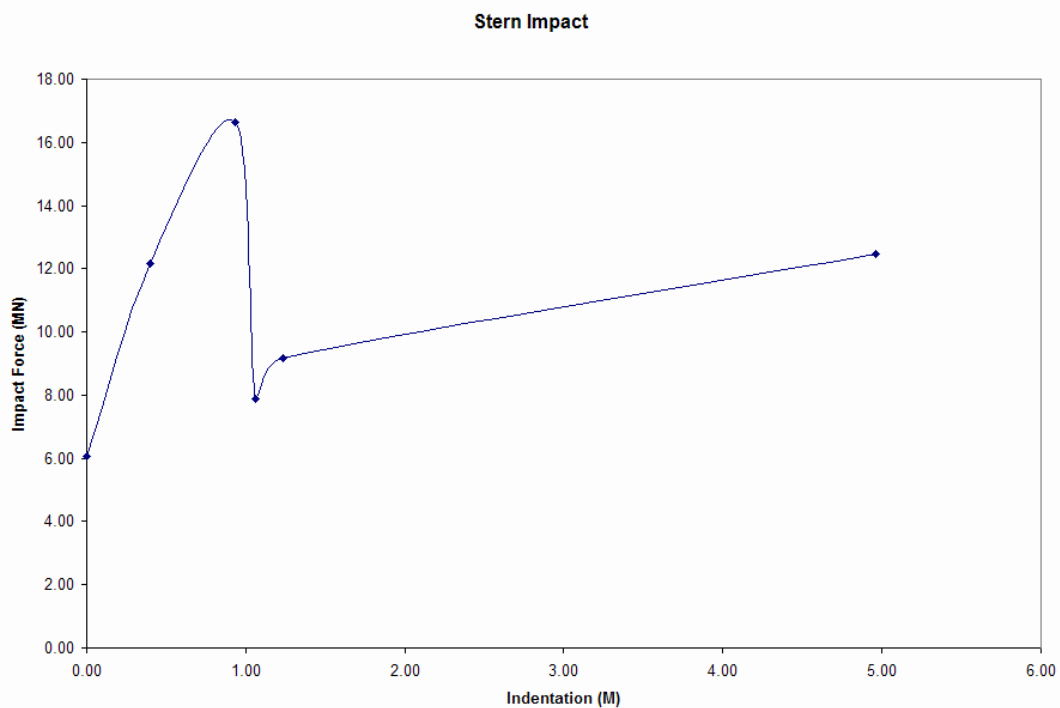
- a. Local deformation of the impacted member due to denting and beam bending.
- b. Global deformation of the entire structure.
- c. Deformation of the ship structure.

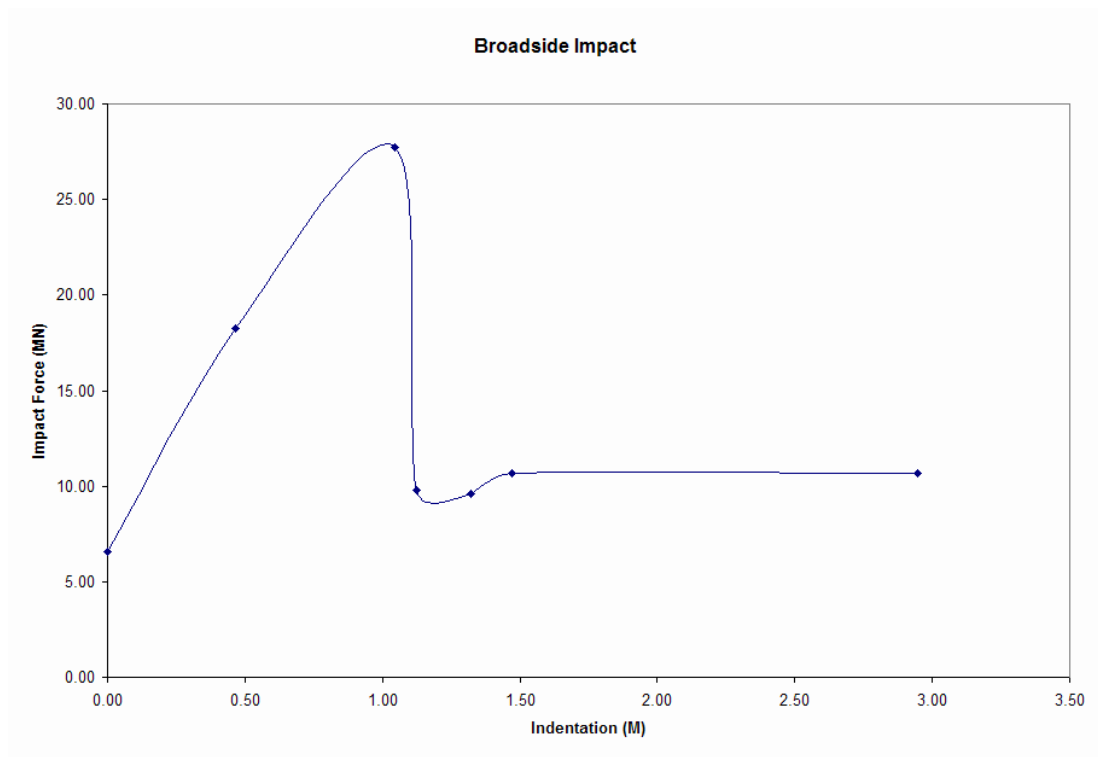
Local deformation of the impacted member due to beam bending and the global deformation of the structure is readily accounted for by Collapse. To account for localized denting it is recommended that the impacted member is modeled using isotropic plate elements. The SACS module Precede has the facility to generate a tubular finite element plate mesh for a given member. Alternatively, the local denting energy of the impacted member may also be taken into account in accordance to either the Ellinas or Furnes approaches outlined in the API RP2A-WSD code of practice by selecting the appropriate option on the IMPACT input line.

NOTE the latter approach does not account for any geometric nonlinearities resulting from local indentations.

A joint force, together with the total kinetic energy or the mass and velocity of the impacting object, can be used to simulate an impact. Collapse allows for automatic unloading for post impact analysis.

Collapse View can be used to produce reports and plots of the energy absorbed by the structure and the ship for both cases where a prescribed displacement or a prescribed force is used to model the impact force. User defined ship indentation curves are available within Collapse together with DNV[1] force displacement curves for a 5000 ton ship and a 1.5m and 10m diameter infinitely stiff cylindrical column similar to the ones shown below. Collapse View has ship indentation curves for 5000 ton ship and 1.5m diameter column and assumes that no more energy is absorbed by the ship once the maximum ship force has been exceeded.





DNV Force - Displacement Curves for a 5000 ton Ship and 1.5m Diameter Column

2 COLLAPSE MODELING AND INPUT

The Collapse program requires a SACS model file and a Collapse input file. The model requires some minor modeling considerations for the purpose of the nonlinear plastic analysis.

2.1 MODELING REQUIREMENTS

A standard SACS model may be used as the model input for the nonlinear analysis with the following requirements:

2.1.1 Analysis Type

The 'NL' analysis type option must be specified in on the model OPTIONS line for standard nonlinear plastic analysis. For nonlinear analysis including a nonlinear elasto-plastic foundation, the 'NP' analysis option must be designated.

2.1.2 Load Combinations

All load cases which are specified as part of a load step in the nonlinear plastic collapse analysis must be basic load conditions. However, because a load sequence may consist of numerous load conditions, any combination of basic load cases can be applied sequentially as part of the load sequence.

Note: Load combinations are accounted for in the Collapse input file by a load sequence consisting of the basic load cases that define the combination applied sequentially. Alternatively, load combinations may be converted to basic load cases using the Seastate program prior to execution of the Collapse analysis.

2.2 COLLAPSE ANALYSIS INPUT

In addition to the model, the nonlinear plastic analysis requires a Collapse input file defining analysis input data.

2.2.1 Collapse Analysis Options

Collapse analysis options are specified in columns 26-41 on the CLOPT line.

2.2.1.1 Joint Flexibility

The effects of tubular connection flexibility may be accounted for by specifying analysis option 'JF'.

Alternatively, participants of the JIP 'Assessment Criteria, Reliability and Reserve Strength of Tubular Joints' may access a formulation for connection flexibility that has been developed by MSL Engineering Limited (UK). The formulation can be specified with analysis option 'MF' for mean level or 'CF' for characteristic level on the input line MSLOPT in columns 8-9.

2.2.1.2 Member Local Buckling

Local buckling of the member cross section may be considered by specifying analysis option 'LB' in one of the analysis options fields. The criteria used for local buckling is specified on columns 52-53 as 'MG'

for Marshall & Gates lower limit of critical strain, '2U' for API Bulletin 2U recommendations or 'LR' for API ultimate strength code criteria.

2.2.1.3 Pile Plasticity

When executing a nonlinear plastic analysis including the pile/soil foundation, the pile elements material properties may be treated as elastic or plastic. Enter 'PP' in one of the analysis option fields to use plastic material properties for pile elements.

2.2.1.4 Considering Skipped Elements Plastically

By default, any element or element group designated in the model file to be skipped for post processing purposes is considered as an elastic element (i.e. have elastic material properties for any step of the nonlinear plastic analysis). Skipped elements may be considered to have plastic material properties by specifying the analysis option 'NS'.

Note: Skipped beam elements are designated in the model file by 'SK' in columns 20-21 on the MEMBER line defining the member or by specifying member class '9' in column 47 on the GRUP line defining the group to which it is assigned. Skipped plates are designated by 'SK' in columns 31-32 on the PLATE line defining it.

2.2.1.5 Tubular Connection Capacity Check

Joint strength check based upon API RP 2A-LRFD recommendations for tubular joints can be implemented by specifying 'JS' in one of the analysis options field between columns 26-41. Alternatively, 'ND' may be specified at the same location in order to perform a joint check based upon the Norsok standard for the design of steel structures.

Once the joint strength check criterion has been exceeded the connection is considered to have failed and the brace stiffness is removed from the analysis.

Alternatively, participants of the JIP 'Assessment Criteria, Reliability and Reserve Strength of Tubular Joints' may access the capacity check that has been developed by MSL Engineering Limited (UK). The capacity check includes mean level and characteristic level options specified with analysis option 'MS' or 'CS', respectively, in columns 10-11 on the MSLOPT line.

2.2.1.6 Strain Hardening

After plasticity occurs in an element, the Collapse program has the ability to include the effects of strain hardening. To consider the effects of strain hardening, enter the strain hardening ratio, defined as the ratio of the slope of the plastic portion of the stress-strain curve to the slope of the elastic portion, in columns 76-80.

2.2.1.7 Collapse Critical Displacement

The collapse critical displacement or the maximum deflection allowed before the structure is considered to be collapsed or failed may be specified in columns 71-75.

2.2.1.8 Creating a SACS Model File at Final Step

A SACS model file with joint coordinates that reflect the final displaced position of the joint may be created by inputting 'SF' in columns 38-39 on the CLPOPT line.

2.2.2 Analysis Parameters and Convergence Criteria

Analysis parameters such as number of plastic member sub-segments and the maximum number of iterations are specified in columns 11-19 on the CLPOPT line while analysis convergence criteria are specified in columns 56-60.

2.2.2.1 Number of Member Sub-Segments

By default, members with plastic material properties are divided into eight sub-segments along the member length. The number of sub-segments for members may be specified in columns 14-16.

Note: The sub-segment length is determined by dividing the total member length by the maximum number of sub-segments designated. For segmented members, any sub-segment which has a change in property is further divided into two constant property sub-segments at the point at which the section property changes. Therefore, segmented members may have more sub-segments than the maximum specified.

2.2.2.2 Member Iterations and Displacement Convergence

For any load increment, a beam-column solution is performed for each plastic member using the cross section sub-element details. Member stiffness iterations continue until the displacements of member sub-segment joints for two successive iterations meet the member displacement tolerance or until the maximum number of member iterations has been met. The default number of member iterations is 20 and may be overridden in columns 17-19. The default member displacement tolerance is 0.01 inch or 0.01cm and may be overridden in columns 66-70.

Note: The maximum number of member iterations may be increased when member solution has not converged.

2.2.2.3 Global Stiffness Iterations and Convergence

For any load increment, a beam-column solution is performed for each plastic member using the cross section sub-element details. The global stiffness iteration is then performed including any effects of connection flexibility and nonlinear pile/soil foundation effects. The deflected shape of the structure is then determined and compared to the displacements of the previous global stiffness iteration. The stiffness iterations are repeated until the displacements and rotations meet the displacement and rotation convergence tolerances or the maximum number of iterations has been met.

By default, the maximum number of global stiffness iterations per load increment is 20 but may be overridden in columns 11-13. The default displacement and rotation convergence tolerances are 0.01 inch or 0.01cm and 0.001 radians and may be overridden in columns 56-60 and 61-65, respectively.

2.2.2.4 *Continue if Maximum Number of Iterations Exceeded*

By default, the nonlinear analysis is terminated when the maximum number of iterations is exceeded. Specify the 'CN' analysis option in one of the analysis options fields, columns 26-41, to continue the analysis even if the maximum number of iterations is exceeded.

2.2.3 Output Reports

Output reports including joint deflections, joint reactions, member internal loads and stresses, collapse summary and member summary reports are available. Report data may generated based on the final analysis results, each load increment or each iteration. Output report options may be specified on the CLPRPT line in columns 8-31.

2.2.3.1 *Joint Displacements*

Joint displacements may be reported for the structure's final position, for each load increment or for each iteration by specifying 'P0', 'P1' or 'P2', respectively.

2.2.3.2 *Selecting Joints for Displacement Report*

By default, the displacements for each joint in the model is reported in the joint displacement report. The user may designate the joints to be reported in the joint displacement report on the JTSEL line. There is no limit to the number of joints that may be designated.

Note: If joints are designated using the JTSEL line, only joints specified are included in the joint displacement report.

2.2.3.3 *Joint Reactions*

Joint reactions may be reported for the structure's final position, for each load increment or for each iteration by specifying 'R0', 'R1' or 'R2', respectively.

2.2.3.4 *Member Internal Loads and Stresses*

Member internal loads and stresses may be reported for the structure's final position, for each load increment or for each iteration by specifying 'M0', 'M1' or 'M2', respectively.

2.2.3.5 *Pilehead Reactions Report*

The pilehead reactions may be reported for the structure's final position, for each load increment or for each iteration by specifying 'F0', 'F1' or 'F2' respectively in columns 26-27 on the CLPRPT input line.

2.2.3.6 *Selecting Members for Internal Loads and Stress Report*

By default, the internal loads and stresses will be reported for all members in the model which can be quiet voluminous. To avoid large reports the user may select specific members to be reported by using the MEMSEL line. There is no limit to the number of members that may be designated.

2.2.3.7 *Selecting Plates for Reports*

By default, reports will be produced for all plates. The user can request reports on specific plates by using the PLTSEL line. There is no limit to the number of plates that may be selected.

2.2.3.8 *Excluding Elastic Members*

Members whose properties remain elastic may be excluded from the internal loads and stress reports by selecting the 'MP' option. The report will thus contain internal loads and stresses only for plastic members.

2.2.3.9 *Designating Minimum Plasticity*

A minimum plasticity ratio for the member stress report may be specified in columns 32-36 on the CLPRPT line. If a minimum plasticity ratio is specified, only members with sub-elements that have plasticity ratios greater than the ratio specified are reported.

2.2.3.10 *Collapse Summary Report*

The Collapse solution summary report containing the load case, load factor, force summation, and maximum displacement and rotation for each load increment may be obtained by specifying report option 'SM'.

2.2.3.11 *Member Summary Report*

Select the 'MS' option to obtain a plastic member summary report including the plasticity ratio and member internal loading for each load increment.

2.2.4 **Applying Load**

Unlike standard linear analysis, the Collapse program analyzes a set of load cases applied step by step or sequentially rather than simultaneously. The Collapse program allows for up to six load sequences to be defined with each load sequence analyzed as an independent nonlinear analysis.

2.2.4.1 *Defining a Load Sequence*

A load sequence defines a set of load steps that will be applied in the sequence or order specified by the user using LDSEQ lines. Enter the load sequence name in columns 7-10 of the first LDSEQ line defining the sequence.

Each load sequence may contain from one to fifty load steps defined in columns 21-80 on the LDSEQ line. A load step defines the basic load case to be applied, the number of increments over which to apply the load case, the initial load case factor and the final load case factor. For any particular load step, the magnitude of each load increment is constant and is determined by:

$$\text{Load Increment} = \frac{(\text{End factor} - \text{Begin factor})}{\text{Number of increments}}$$

Note: The order in which loading is applied in the sequence may have a significant effect on the analysis results. For example, dead loading or self weight should be applied before any environmental loading.

2.2.4.2 *Load Sequences with More than Three Load Steps*

Multiple LDSEQ lines may be used to define load sequences consisting of more than three load steps. For each subsequent LDSEQ line, leave the load sequence ID in columns 7-10 blank to designate that the

load steps defined are a continuation of the current load sequence. Up to a total of seventeen LDSEQ lines may be used to define up to fifty steps for any particular load sequence.

2.2.4.3 Using Load Combinations

Although only basic load cases may be specified as part of a load sequence, load combinations may be analyzed by defining the basic load cases making up the combination, as part of the load sequence. Unlike linear analysis, these basic load conditions are applied sequentially rather than simultaneously.

Alternatively, load combinations may be converted to basic load cases using the Seastate program prior to execution of the Collapse analysis.

2.2.5 Tubular Connection Capacity Parameters

2.2.5.1 Tubular Connection Capacity Options

Joint strength options used for the tubular connection capacity check can be implemented through the use of the JSOPT line. This line is optional in any collapse analysis. If this line is omitted then default options will be used.

2.2.5.2 LRFD Resistance Factor Data

By default, the Collapse program will use the LRFD safety indices that are specified in the API RP 2A-LRFD commentary as resistance factors. Alternative resistance factors can be implemented by the use of the RSFAC input line.

2.2.5.3 Norsok Resistance Factor Data

Resistance factors may be used in conjunction with the Norsok joint strength check. Connection and material resistance factors default to 1.0 and 1.15 respectively. Alternative resistance factors can be specified by the use of the RSFAC input line.

2.2.6 Designating Elements as Elastic

By default, members and groups designated as skipped for post processing are treated as large deflection elements with elastic material properties. Additionally, members or member groups may be designated by the user as elastic elements using the MEMELA and GRPELA input lines, respectively. Similarly, plate elements and plate groups can be designated as elastic elements using the PLTELA and PGRELA input lines respectively.

Note: Designating elements to remain elastic can significantly reduce the run time for a collapse analysis. Also, certain element types including wishbones, non-structural framing, i.e. framing representing risers, boat landings, anodes, etc. and dummy framing should be treated as elastic elements for the purpose of the nonlinear analysis.

2.2.6.1 Elastic Members

Specify the start and begin joints of any member that is to be considered as a large deflection elastic element on the MEMELA input lines. As many MEMELA lines as required may be specified.

2.2.6.2 Elastic Member Groups

Specify member groups to which all elements assigned are to be considered as large deflection elastic elements on the GRPELA input line. As many GRPELA lines as required may be specified.

2.2.6.3 Elastic Plates Elements

Specify the plate ID's of plate elements that are to be considered as large deflection elastic elements on the PLTELA input lines. As many PLTELA lines as required may be specified.

2.2.6.4 Elastic Plate Groups

Specify plate group names that are to be considered as large deflection elastic elements on the PGRELA input line. As many PGRELA lines as required may be specified.

2.2.7 Nonlinear Springs

The Collapse program supports nonlinear springs and nonlinear spring supports.

2.2.7.1 Nonlinear Spring Supports

A general nonlinear spring to ground element is available in Collapse. The spring elements have six uncoupled degrees of freedom. The force deflection characteristics of the spring for each degree of freedom are defined by discrete Force-Displacement points in the input line NLSPRG. Up to four points may be used to define the spring Force-Displacement characteristics. As many NLSPRG input lines as required may be specified.

2.2.7.2 Joint to Joint Nonlinear Springs

Nonlinear springs can be assigned between existing joints. The force deflection characteristics of the spring for each degree of freedom are defined by discrete Force-Displacement points in the input line NLSPJJ. As many points as required may be used to define the spring Force-Displacement characteristics. As many NLSPJJ input lines as required may be specified.

2.2.8 MSL Joint Flexibility Formulation

Participants of the joint industry project 'Assessment Criteria, Reliability and Reserve Strength of Tubular Joints' may access the joint flexibility formulation developed by MSL Engineering Limited (UK). Options from the formulation may be accessed on the MSLOPT line.

Two levels of tubular connection capacity, 'mean' level and 'characteristic' level are included. The 'mean' level corresponds to a 50% probability of survival while the 'characteristic' level corresponds to a 95% probability of survival.

2.2.8.1 Joint Flexibility

The predicted effects of tubular connection flexibility may be accounted for by specifying analysis option 'MF' or 'CF' for mean or characteristic level, respectively, in columns 8-9.

By default, a convergence tolerance of 0.001 is assumed for joint distortion and rotation. The joint distortion tolerance can be specified in columns 15-19. The joint rotation tolerance can be specified in columns 20-24.

2.2.8.2 Joint Strength

The predicted tubular connection strength at 'mean' level can be accounted for by specifying analysis option 'MS' in columns 10-11. Alternatively, the connection strength may be assessed at the characteristic level by specifying 'CS' in columns 10-11.

2.2.8.3 Fracture Criteria

The ductility limits for tension loaded joints may be accounted for by specifying analysis option 'MT' at mean level, and 'CT' at characteristic level in columns 12-13.

2.2.9 Joint Strength/Flexibility Selection

Individual joints may be chosen for joint strength or joint flexibility analysis. The option used, either joint strength 'JS' or joint flexibility 'JF', must be specified with CLPOPT analysis options. With the 'JS' option specified on the CLPOPT line, a joint or group of joints may be chosen for joint strength analysis with the JSSEL line. This means that all braces connected to the joints specified will be included or excluded from the joint strength analysis. The line either includes or excludes the joints specified in columns 9-77 based on the entry in column 7. Specifying 'I' in column 7 will mean that the joints named are included in the joint strength analysis; specifying 'X' in column 7 will mean that all joints except those named are included in the joint strength analysis.

In the same manner, joints may be chosen for joint flexibility analysis with the JFSEL line. With either JSSEL or JFSEL, the include or exclude option is mutually exclusive. Therefore, if multiple lines are used to include or exclude joints, each line must have the same option specified in column 7.

In the following example, joints 101 and 102 are excluded from joint flexibility analysis. All other joints will be analyzed.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	JFSEL	X	101	102				

If the choice of a single joint for joint strength or joint flexibility analysis is not sufficiently restrictive, the BSSEL and BFSEL allow the user to restrict strength or flexibility analysis to individual brace/chord connections. The option used, either joint strength 'JS' or joint flexibility 'JF', must be specified with CLPOPT analysis options. With the 'JS' option specified on the CLPOPT line, a brace/chord connection joint may be chosen for joint strength analysis with the BSSEL line. The first brace member joints are specified in columns 9-12 (begin joint) and columns 13-16 (end joint). The strength analysis will be calculated at the brace/chord connection joint, which is either the begin joint or the end joint of the brace member, and is specified in columns 17-20 for the first brace. Up to five braces may be specified on the BSSEL line. As in the JSSEL line, brace/chord connections may be included or excluded from strength analysis by specifying 'I' or 'X' in column 7.

Equivalently, joint flexibility for individual brace/chord connections is specified with the BFSEL line. With either BSSEL or BFSEL, the include or exclude option is mutually exclusive. Therefore, if multiple lines are used to include or exclude brace/chord connection joints, each line must have the same option specified in column 7.

In the following example, brace/chord connection joint 101 of brace member 101-401 is excluded from brace strength analysis. All other brace/chord connections will be analyzed.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	BSSEL	X	101	401	101			

The resistance factor specified for a brace/chord connection may be modified using the RSFACO line. This line allows the user to override joint resistance factor values specified on RSFAC lines. The line specifies the brace member in columns 8-11 (begin joint) and columns 12-15 (end joint). The brace/chord connection joint, which is either the begin joint or the end joint, is specified in columns 16-19. The resistance factors (axial tension, axial compression, in-plane bending, out-of-plane bending, yield stress) are specified in columns 21-45. Optionally, the connection type may be specified in column 47, with choices being 'X' (X or cross connection), 'Y' (T or Y connection), or 'K' (K brace connection). Any of the resistance factors left unspecified or given the value 0.0 will be replaced by values specified for the connection joint on previous RSFAC lines.

In the following example, brace/chord connection joint 201 of brace member 201-501 will have an in-plane bending resistance factor of 3.81 and an out-of-plane resistance factor of 3.61. The values for the axial tension, axial compression and yield stress resistance factors are the values specified earlier on RSFAC lines for joint 201.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	RSFACO	201	501	201	3.81	3.61		

2.2.10 Material Properties

Detailed multilinear post yield stress-strain behavior for members can be specified using this option. If this option is not selected, post-yield behavior remains linear and is governed by the strain hardening ratio specified in the CLPOPT line. For this purpose a set of material models have to be defined and different member groups can be assigned to each of these material models. MATGRP line is used to assign a material model to several member groups. Plastic strain (defined as total strain minus yield strain) and Stress factor (defined as actual stress divided by the yield stress) values beginning at the yield point are entered in MATPRP PLAS lines to define the post-yield behavior of the material.

Note 1: This option is available only for member elements. This functionality will be extended to piles and plate elements in future.

Note 2: The plastic strain values and the stress factor values should be in the monotonically increasing order.

Note 3: Maximum of 50 different material models and maximum of 50 data points (1 strain-stress couple =1 data point) in each material model are permitted.

Note 4: For calculating the stress value corresponding to a plastic strain value greater than the maximum value specified in the material model (beyond the last data point in the material model), the constant strain hardening ratio specified in CLPOPT line will be used.

The following table shows a sample of how to calculate the Plastic Strain and Stress Factor values from the actual stress-strain curve of a hypothetical material:

$\sigma_y = 144.9 \text{ N/mm}^2$; $E = 1.4 \times 10^5 \text{ N/mm}^2$; Therefore, $\epsilon_y = (\sigma_y/E) = 0.00104$			
Actual Stress-Strain Curve		Plastic Strain-Stress Ratio Curve	
Strain (ϵ)	Stress (σ)	Plastic Strain ($\epsilon_p = \epsilon - \epsilon_y$)	Stress Ratio (σ/σ_y)
0.0000	0.0		
0.00104 (ϵ_y)	144.9 (σ_y)	0.00000	1.000
0.00262	215.7	0.00158	1.489
0.00420	248.5	0.00316	1.715
0.00736	290.0	0.00632	2.001
0.01052	316.0	0.00948	2.181
0.01368	332.6	0.01264	2.296
0.01684	342.0	0.01580	2.360
0.02000	345.0	0.01897	2.381

Yield Point →

The following image shows how this data will be entered in the Collapse input file. The MATGRP lines are used to assign MAT1 material model to 19 different member groups (G01 to G19). MATGRP lines are immediately followed by a MATPRP HEAD line and a number of MATPRP PLAS lines to define the material model MAT1. To add another material model, these set of lines should be repeated.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	MATGRP MAT1	G01 G02 G03 G04 G05 G06 G07 G08 G09 G10 G11 G12 G13 G14 G15						
2	MATGRP	G16 G17 G18 G19						
3	*							
4	MATPRP HEAD MAT1							
5	*							
6	MATPRP PLAS	0. 1.0.00158 1.489.00316 1.715.00632 2.001.00948 2.181						
7	MATPRP PLAS	.01264 2.296.01580 2.360.01897 2.381						

3 TROUBLE SHOOTING

3.1 MODEL SINGULARITY

Model singularity is the common term used to describe problems within a stiffness matrix that may limit the accuracy of a solution or prevent it entirely. In matrix theory, a structural model matrix must be positive definite for it to be inverted. Some common problems for a matrix to be non-positive definite are as follows:

1. Portion of structure or entire structure translating or rotating as a rigid body in space.
2. A joint connected to the structure is translating or rotating in space because a particular end fixity for all members connecting to the joint is released.
3. Member or plate structural properties are zero for all elements connected to a joint so that the joint is effectively unrestrained.
4. The structural stiffness is negative due to structural collapse through the occurrence of a mechanism. This may occur due to insufficient strain hardening.

3.2 DEBUGGING THE MODEL

If the Collapse program detects a non-positive definite diagonal term in the stiffness matrix, the row of the matrix where it occurred will be indicated. If the value is between zero and -0.0001 it will be reset to 1.0 and the row and column where it occurred will be nulled and solution will continue. If the diagonal value is less than -0.0001 the program terminates execution and reports the critical joint degree of freedom.

For instances where an unrestrained portion of the structure acts as a mechanism for a singularity to occur, the last joint of the mechanism, in optimized order is reported. If the reported joint is indeed unrestrained, the Interpreted Input Echo Report can be used to isolate the critical portion of the structure. The interpreted Joint Data List portion of the report contains the joint degree of freedom and matrix row location list in the following format:

1. The degree of freedom for each joint in the stiffness matrix as rotation X, Y and Z followed by translation X, Y and Z.
2. For each joint, the beginning row number corresponding to the rotation X degree of freedom is listed in the report. The row numbers corresponding to rotation Y, Z and translation X, Y and Z are obtained by adding 1, 2, 3, 4 to the joint rotation X degree of freedom.

The critical row location is reported in the solution listing file.

3.3 WARNING MESSAGES IN COLLAPSE

3.3.1 Non-convergence of Piles

```
*** ERROR-MAX. ITERATIONS EXCEEDED AT PILE JOINT 'joint name'
```

This error message occurs when the procedure used to calculate the stiffness and plasticity of a pile has failed to converge. The specific pile that has caused the problem is attached to the joint specified by 'joint name'.

The determination of the stiffness and plasticity of a single pile requires the solution of a nonlinear problem which may involve a number of iterations. The convergence of this procedure is governed by the displacement convergence requirement, which is specified on the PSIOPT line of the PSI input file that is used for the analysis.

The maximum number of iterations that are used to solve for each pile is 100. If convergence has not taken place prior to the 100th pile iteration, then the error message (above) is displayed in the Collapse listing file, and the pile solution process is terminated. Subsequently, two informational messages are displayed containing data that are related to components of force and deflection at the pilehead.

```
*** FORCES - ***** ...
```

```
*** DEFLECTIONS - ***** ...
```

These messages do not contain useful information and it is recommended that they be ignored.

Two likely causes of pile non-convergence are:

1. A tight displacement convergence requirement.
2. Instability in the supported structure.

In the case of item 2, it is suggested that a run be made without piles in order to assess if the supported structure is stable.

3.3.2 Maximum Allowable Displacement or Rotation

```
**** WARNING - EXCEEDED MAXIMUM ALLOWED DISPLACEMENT OR ROTATION
```

This warning message occurs on completion of a load increment if the deflection of any joint's degree-of-freedom exceeds a prescribed limit. For degrees-of-freedom that allow translation, the default maximum deflection is 1000.0 in. (393.7 cm.). However, the user can specify a translational limit directly by using the 'Collapse Deflection' field in columns 71-75 of the CLPOPT line. The deflection is specified in units consistent with those of the SACS system configuration.

There is also a limit for rotational degrees of freedom, which is set to 2.0 radians. On detection of a displacement or rotation having been exceeded, the following warning message is displayed and the analysis is terminated.

```
**** WARNING - STRUCTURE COLLAPSED *****
```

3.3.3 Non-convergence of a Load Increment

```
**** WARNING - EXCEEDED MAXIMUM ITERATIONS OF 'N'
```

Where 'N' is a user-specified value that represents the maximum number of iterations for a given load increment.

This warning message occurs if the maximum number of iterations has been exceeded for a given load increment. The maximum number of iterations per increment should be specified by the user in columns 11-13 of the CLPOPT line. If no user specification is made, then the maximum number defaults to 20.

Collapse will attempt to use a sufficient number of iterations to achieve convergence for a given load increment. However, if Collapse attempts to use a number of iterations that is greater than the maximum, no further iterations are performed for the current load increment, and the analysis is declared to be non-converged for that increment.

By default, if the number of iterations has been exceeded for a load increment, the analysis will terminate and the warning message will be displayed in the Collapse listing file. However, if the user has specified 'CN' in columns 28-29 on the CLPOPT line, the analysis will continue with the next load step after the warning message has been displayed.

Non-convergence due to the requirement for a large number of iterations is often associated with, but not limited to, the following circumstances:

1. One or more of the convergence tolerances on the CLPOPT line is tight.
2. A low strain hardening ratio.
3. A portion of the load step has approached an unstable region brought about by the failure of an entity such as a joint or a member.
4. The effective incremental stiffness of an element is almost zero.

In the event of non-convergence of a load increment, it is suggested that the maximum number of iterations be increased from 20 to 40. Increasing the maximum number of iterations beyond 40 does not normally improve convergence.

3.3.4 Non-convergence of Members

*** WARNING - ELEMENTAL STIFFNESS FOR MEMBER ***** NOT CONVERGED

This warning message occurs when the procedure used to calculate the stiffness and plasticity of a member has failed to converge. The message is only displayed if member warning messages have been enabled by specifying 'PW' in columns 24-25 of the CLPRPT line.

The determination of the stiffness and plasticity of a single member requires the solution of a nonlinear problem which may involve a number of iterations. The convergence of this procedure is governed by the member deflection tolerance, which is specified in columns 66-70 of the CLPOPT line.

The maximum number of iterations that are used to solve for each member can be specified by the user in columns 17-19 of the CLPOPT line. The default maximum number of member iterations is 20. If convergence has not taken place prior to the maximum allowable member iteration, then the error message (above) is displayed in the Collapse listing file, and the member solution process is terminated. Subsequently, three informational messages are displayed containing data that are related to force and convergence criteria.

ERR =***** ...

ALLOWABLE= ***** ...

FAXIAL= ***** ...

These messages do not contain useful information and it is recommended that they be ignored.

Likely causes of member non-convergence include:

1. A tight member deflection tolerance.
2. The maximum number of member iterations is too small.
3. The member has become very deformed.
4. Instability in the rest of the structure.

4 COMMENTARY

4.1 ENERGY PRINCIPLES

The energy, or variational methods of structural mechanics constitute a powerful and widely used approach. Forms of these methods have been tools for the analysis of engineering structures for more than a century. The application of energy methods to the derivation of forces and displacements in a structure was developed by Castiglino^[1] in the 1870s. The application of complementary energy for the analysis of nonlinear structures was developed by Engesser^[1] in 1889. Since then a number of theorems have been formulated on the bases of these developments. The following section discusses the basic energy variational principles employed by the nonlinear Collapse program.

4.1.1 Discrete Systems

Consider a discrete system where the potential energy, V can be expressed as function of displacements q_i and loads P_i . If the system is subjected to a small variation in displacements δq_i , so that its new configuration is $q_i + \delta q_i$ (assuming load P_i remains constant), the potential energy of the system in its new configuration can be expressed via a Taylor's series expansion as:

$$\begin{aligned} V(P_i, q_i + \delta q_i) &= V(P_i, q_i) + \frac{\partial}{\partial q_i} V(P_i, q_i) \delta q_i \\ &+ \frac{1}{2!} \frac{\partial^2}{\partial q_i \partial q_j} V(P_i, q_i) \delta q_i \delta q_j + \dots \end{aligned} \quad (1)$$

Equation (1) can be written in a simplified form as:

$$\delta^T V = \delta V + \delta^2 V + \dots \quad (2)$$

Where $\delta^T V$ is the total variation in the potential energy expressed by:

$$\delta^T V = V(P_i, q_i + \delta q_i) - V(P_i, q_i) \quad (3)$$

δV and $\delta^2 V$ are the first and second variations of the potential energies given by:

$$\delta V = \frac{\partial}{\partial q_i} V(P_i, q_i) \delta q_i \quad (4)$$

And

$$\delta^2 V = \frac{1}{2!} \frac{\partial^2}{\partial q_i \partial q_j} V(P_i, q_i) \delta q_i \delta q_j \quad (5)$$

4.1.1.1 Discrete System - Equilibrium

For a system to be in equilibrium, the potential energy is stationary with respect to displacements so that for all admissible values of δq_i , the first variation of the total potential energy is zero, i.e.:

$$\delta V = \frac{\partial}{\partial q_i} V(P_i, q_i) \delta q_i = 0 \quad (6)$$

Equation (6) yields n equilibrium equations for $i = 1, n$. If the system is considered to be in an equilibrium configuration so that $\delta V = 0$, equation (2) may be rewritten as:

$$\delta^T V = \delta^2 V + \dots \quad (7)$$

4.1.1.2 Discrete System - Unstable Equilibrium

If the system in its new configuration is in a state of stable equilibrium, then the total variation in potential energy, $\delta^T V$, is a minimum and the second variation $\delta^2 V$ is a quadratic form in dq_i and is positive definite for all admissible values of δq_i . Unstable or critical conditions occur when $\delta^2 V$ changes from positive definite to semi-positive definite indicating a possible transition from stable equilibrium to unstable equilibrium.^[2]

4.1.1.3 Discrete System - Nonlinear Problems

For nonlinear problems, the first variation of the potential energy, δV , yields n unknown nonlinear equations in the displacement variables q_i ($i = 1, n$).

If Δ denotes a small but finite increment in displacements and forces, then expanding the first variation of the potential energy $\delta V(P_i + \Delta P_i, q_i + \Delta q_i)$ in a Taylor series about the (P_i, q_i) configuration yields:

$$\begin{aligned} \delta V(P_i + \Delta P_i, q_i + \Delta q_i) &= \delta V(P_i, q_i) + \frac{\partial}{\partial P_i} V(P_i, q_i) \Delta P_i \\ &\quad + \frac{\partial}{\partial q_i} V(P_i, q_i) \Delta q_i + \dots \end{aligned} \quad (8)$$

Rearranging equation (8) and retaining only first order terms in increments Δ yields:

$$\begin{aligned} \delta V(P_i + \Delta P_i, q_i + \Delta q_i) - \delta V(P_i, q_i) &= \frac{\partial}{\partial P_i} V(P_i, q_i) \Delta P_i \\ &\quad + \frac{\partial}{\partial q_i} V(P_i, q_i) \Delta q_i + \dots \end{aligned} \quad (9)$$

If the system in configuration $(P_i + \Delta P_i, q_i + \Delta q_i)$ is in equilibrium then:

$$\delta V(P_i + \Delta P_i, q_i + \Delta q_i) = 0 \quad (10)$$

Substituting equation (10) into equation (9), rearranging the terms and ignoring higher order terms yields the following equation:

$$\frac{\partial}{\partial q_i} V(P_i, q_i) \Delta q_i = -\frac{\partial}{\partial P_i} V(P_i, q_i) \Delta P_i - \delta V(P_i, q_i) \quad (11)$$

Equation (11) provides a basis for an iterative procedure for the solution of nonlinear equilibrium equations. If the second term, $\delta V(P_i, q_i)$, is set to zero, then equation (11) represents the incremental equations of equilibrium.

4.1.2 Continuous Systems

The variational principles for discrete systems can be extended to continuous systems.^[4] The loads P_i and displacements q_i in the discrete system can be assumed analogous to the externally applied loads and nodal displacement coefficients which define the magnitude of displacements in continuous systems.

4.1.2.1 Continuous Systems - Equilibrium

For a system comprising a deformable body acted upon by external forces P_i with the corresponding displacements defined by r_i , the first variation of the potential energy is zero when the system is in a state of equilibrium. Assuming that the external forces remain constant, this can be represented by the following equation: ^[4]

$$\delta V = -P_i \delta r_i + \int \sigma_i \delta \epsilon_i dVOL = 0 \quad (12)$$

where the repeated suffices imply summation, σ_i represents the internal stresses, $\delta \epsilon_i$ represents the first variation in the corresponding strains and the integration is over the volume of the body.

Noting that $\delta^2 V = \delta (\delta V)$, the second variation of V is given by ^[4]:

$$\int (\sigma'_i \delta \epsilon_i + \sigma_i \delta \epsilon'_i) dVOL = P'_i \delta r_i + P_i \delta r_i - \int \sigma_i \delta \epsilon_i dVOL \quad (13)$$

4.1.2.2 Continuous Systems - Unstable Equilibrium

For stable equilibrium, the first variation corresponds to a minimum and is zero and the second variation is positive definite for all variations in displacements. Unstable or critical conditions occur when $\delta^2 V$ changes from positive definite to semi-positive definite.

Note: Because the second variation of any linear function vanishes, it is necessary to consider second order strains and displacements to completely define equation (13).

4.1.2.3 Continuous Systems - Nonlinear Problems

Assuming that σ_i and ϵ_i can be expressed as functions of displacement variables and r_i can be expressed as a linear function of displacement variables, equations (11) and (12) yield:

$$\int (\sigma'_i \delta \epsilon_i + \sigma_i \delta \epsilon'_i) dVOL = P'_i \delta r_i + P_i \delta r_i - \int \sigma_i \delta \epsilon_i dVOL \quad (14)$$

in which the prime implies the operation

$$\frac{\partial f(x)}{\partial x_i} \Delta x_i \quad (14a)$$

with respect to the applied load and displacement variables ^[5].

Equation (14) is analogous to equation (11) for a discrete system and provides a bases for an iterative procedure for the analysis of nonlinear equilibrium equations. If the last two terms on the right hand side of the equation are set to zero, equation (14) represents the incremental equations of equilibrium.

4.2 NON-LINEAR PLASTIC FORCE APPROACH

For an elasto-plastic problem, the strains can be represented in terms of the displacement variables in matrix form as:

$$\{\epsilon_i\} = [A]\{q\} \quad (15)$$

where ϵ_i is the total strain vector at a point and can be composed of the elastic strains $\epsilon_{e,i}$ and the plastic strains $\epsilon_{p,i}$ so that:

$$\{\epsilon_i\} = \{\epsilon_{e,i}\} + \{\epsilon_{p,i}\} \quad (16)$$

Stresses σ_i which are only dependent upon elastic strains can be expressed as:

$$\{\sigma_i\} = [E]\{\epsilon_{e,i}\} \quad (17)$$

Noting that:

$$\{\sigma'_i\} = [E]\{\epsilon'_i\} \quad (18)$$

And substituting equations (16) - (18) into equation (14) gives:

$$\begin{aligned}
& \int \left([\{\delta q\}^T [A]^T - \{\delta \varepsilon_p\}^T] [E] [[A]\{\Delta q\} - \{\varepsilon'_p\}] + [E] [[A]\{q\} - \{\varepsilon_p\}] \right) dVOL \\
& = \{\delta q\}^T \{\Delta P\} + \{\delta q\}^T \{P\} - \int \left([\{\delta q\}^T [A]^T - \{\delta \varepsilon_p\}^T] [E] [[A]\{q\} - \{\varepsilon_p\}] \right) dVOL
\end{aligned} \tag{19}$$

Where

$$\{P\} = \{P_a\} + \{P_p\} \tag{20}$$

$$\{\Delta P\} = \{\Delta P_a\} + \{\Delta P_p\} \tag{21}$$

$\{P_a\}$ is the applied load vector and $\{P_p\}$ is the plastic load vector and $\{\Delta P_a\}$ and $\{\Delta P_p\}$ are the corresponding load increment vectors.

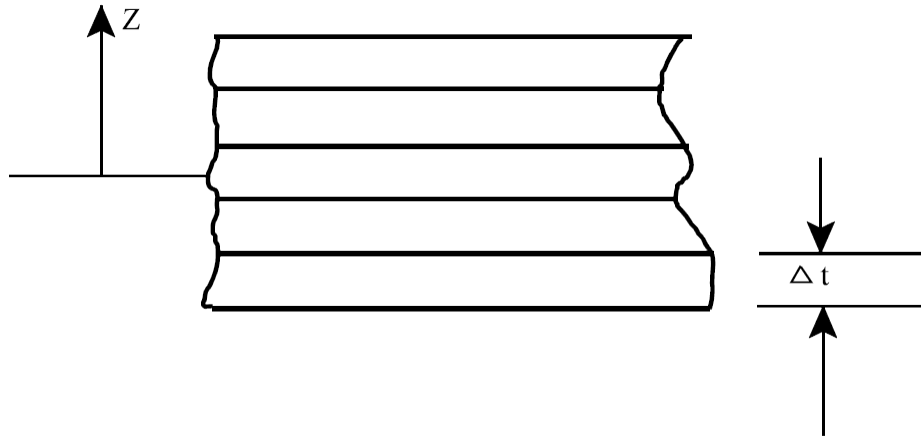
Since the degree of plasticity incurred (and consequently the plastic load vector) is a function of the load path, the solution of an elasto-plastic problem must be handled on an incremental basis given by equation (19) which represents a set of linear simultaneous equations in the unknowns $\{\Delta q\}$ and $\{\Delta P_p\}$. The solution procedure involves the application of a linear load increment $\{\Delta P\}$, and solving the equations for the unknown increments. The improved approximations of $q + \Delta q$ and $P + \Delta P$ are then used as a starting point for the next improvement cycle. The procedure is continued until equilibrium is satisfied, as evidenced by the vanishing of the last two terms on the right hand side of equation (19).

4.3 PLATE ELEMENTS

Thin plates are often used as structural components since they can sustain loads well in excess of their elastic buckling load. The elastic buckling load of such elements has little or no effect on predicting the failure load. At the onset of elastic buckling, the plate behavior becomes nonlinear and the collapse load is normally associated with plastic failure. Elastic buckling may be precluded altogether for thick plated structures where the collapse load is reached through the onset plastic failure. Therefore, when analyzing such structures, it is necessary to include both geometric and material nonlinearities.

There are two main approaches to the elasto-plastic analysis of plates.^[6] The first method, the Area approach, is an approximate approach which assumes sudden plastification of the entire plate thickness as soon as the extreme fiber stress reaches yield. The second approach allows for a gradual plastification through the thickness of the plate by monitoring the stresses at various sub-layers through the plate cross section.

The Collapse program utilizes the second approach where the plate is divided into 5 sub-layers through its thickness as shown below.



Geometric nonlinearities are included through the use of the second order membrane strain expressions given below:^[7]

$$\epsilon_{mx} = \frac{\partial u}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2 \quad (22)$$

$$\epsilon_{my} = \frac{\partial v}{\partial y} + \frac{1}{2} \left(\frac{\partial w}{\partial y} \right)^2 \quad (23)$$

$$\gamma_{mxy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \quad (24)$$

Where ϵ_{mx} and ϵ_{my} represent the membrane strains in the x and y directions, respectively, and γ_{mxy} represents the membrane shear strain.

Bending strain is represented by the following expressions:

$$\epsilon_{bx}} = -z \frac{\partial^2 w}{\partial x^2} \quad (25)$$

$$\epsilon_{by} = -z \frac{\partial^2 w}{\partial y^2} \quad (26)$$

$$\gamma_{bxy} = -2z \frac{\partial^2 w}{\partial x \partial y} \quad (27)$$

Where ϵ_{bx} and ϵ_{by} are the bending strains in x and y directions, respectively, and γ_{bxy} is the bending strain due to twisting.

For an isotropic elastic material, the stress vector $\{\sigma\}^T = \{\sigma_x, \sigma_y, \tau_{xy}\}$ and can be related to the strains through equation (17) as shown below:

$$\{\sigma_i\} = [E]\{\epsilon_i\} \quad (28)$$

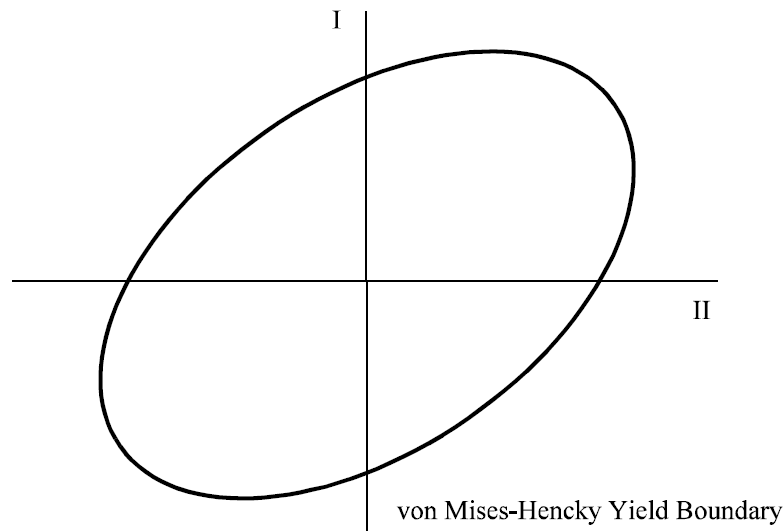
The incremental form of equation (28) is given by:

$$\{\Delta\sigma_i\} = [E]\{\Delta\epsilon_i\} \quad (29)$$

Using the above expressions and utilizing equation (19), it is possible to conduct an elasto-plastic analysis of plated structures. The stresses are monitored at each sub-layer through-out the loading history. The von Mises-Hencky Yield Criterion^[8] is used to determine the onset of plasticity at any sub-layer using the following equation:

$$\Xi = \sigma_x^2 + \sigma_y^2 - \sigma_x \sigma_y + 3\tau_{xy}^2 - S_y^2 = 0 \quad (30)$$

Which defines the yield surface as shown below:



When Ξ is greater than zero, the direction of the plastic strain vector is established by the plastic flow rule according to the theory of plastic potential. Once the plastic strains are determined, the plastic stresses become:

$$\{\Delta\sigma_p\} = [E]\{\Delta\epsilon_p\} \quad (31)$$

The plastic stress resultants are obtained by summing across the plate thickness as follows:

$$N_x = \sum_{i=1}^n \sigma_{xpi} \Delta t \quad M_{xx} = \sum_{i=1}^n \sigma_{xpi} z_i \Delta t \quad (32)$$

$$N_y = \sum_{i=1}^n \sigma_{ypi} \Delta t \quad M_{yy} = \sum_{i=1}^n \sigma_{ypi} z_i \Delta t \quad (33)$$

$$N_{xy} = \sum_{i=1}^n \tau_{xypi} \Delta t \quad M_{xy} = \sum_{i=1}^n \tau_{xypi} z_i \Delta t \quad (34)$$

The plastic nodal force vector for the plate is determined once plate stress resultants are acquired. The plastic nodal force vector is transformed into the global coordinates and added to the global plastic force vector.

4.4 BEAM ELEMENTS

4.4.1 Nonlinear Strain Expressions

The complete nonlinear expressions^[4] for the strains occurring in the tubular, wide flange, angles, channels and tee cross section types is given by the following equation:

$$\begin{aligned} \epsilon_x = & -y \left(\frac{d^2 v}{dx^2} + \frac{d^2 w}{dx^2} \theta \right) - z \left(\frac{d^2 w}{dx^2} + \frac{d^2 v}{dx^2} \theta \right) \\ & + \alpha_w \left(\frac{d^2 \theta}{dx^2} - \frac{d^3 w}{dx^3} \frac{dv}{dx} + \frac{d^3 v}{dx^3} \frac{dw}{dx} \right) \\ & + \frac{du}{dx} + \frac{1}{2} \left(\frac{du}{dx} \right)^2 + \frac{1}{2} \left(\frac{dv}{dx} \right)^2 + \frac{1}{2} \left(\frac{dw}{dx} \right)^2 + \frac{\rho^2}{2} \left(\frac{d\theta}{dx} \right)^2 \end{aligned} \quad (35)$$

The first two terms in the above equation represent the bending strains including the interaction between bending and twisting. The terms on the last line of the equation represent strains produced by stretching of an element due to displacements u, v and w. The third or middle term in the expression results from the restraint in warping. In practice, partial or no restraint in warping may exist and may differ for various structural connection types. Because of this, it is difficult to quantify and hence is not considered by the program. The second order strain in u can also be neglected in the above equation since its contribution can be assumed to be small in comparison with other terms. This results in the following strain expression:

$$\begin{aligned}\epsilon_x = & -y\left(\frac{d^2v}{dx^2} + \frac{d^2w}{dx^2}\theta\right) - z\left(\frac{d^2w}{dx^2} + \frac{d^2v}{dx^2}\theta\right) \\ & + \frac{du}{dx} + \frac{1}{2}\left(\frac{dv}{dx}\right)^2 + \frac{1}{2}\left(\frac{dw}{dx}\right)^2 + \frac{\rho^2}{2}\left(\frac{d\theta}{dx}\right)^2\end{aligned}\quad (36)$$

The expression for shear strain due to St. Venant torsion [9] is given by the following expression:^[4]

$$\epsilon_{xy} = 2\xi\left(\frac{d\theta}{dx} - \frac{d^2w}{dx^2}\frac{dv}{dx} + \frac{d^2v}{dx^2}\frac{dw}{dx}\right)\quad (37)$$

When considering the effects of St. Venant torsion on thin walled bars of open cross section, the section can be considered to be composed of single or several disconnected rectangular strips.

4.4.2 Nonlinear Problems

For a thin walled bar of open cross-section, the first variation of the total potential energy δV is given by:^[10]

$$\delta V = -P_i \delta r_i + \int (\sigma_x \delta \epsilon_x + \sigma_{xy} \delta \epsilon_{xy}) dVOL\quad (38)$$

where σ_x is the axial stress (tensile positive), $\delta \epsilon_x$ is the first variation of the axial strains, ϵ_{xy} is the shear stress and $\delta \epsilon_{xy}$ is the first variation in the corresponding strain. The relationship between the stresses and strains may be given by:

$$\sigma_x = E \epsilon_x\quad (39)$$

$$\sigma_{xy} = G \epsilon_{xy}\quad (40)$$

where E is the Young's Modulus and G is the shear modulus.

Equation (14) provides a basis for an iterative procedure to the solution of nonlinear equations. For a thin walled bar of open cross-section, equation (14) can be rewritten as:

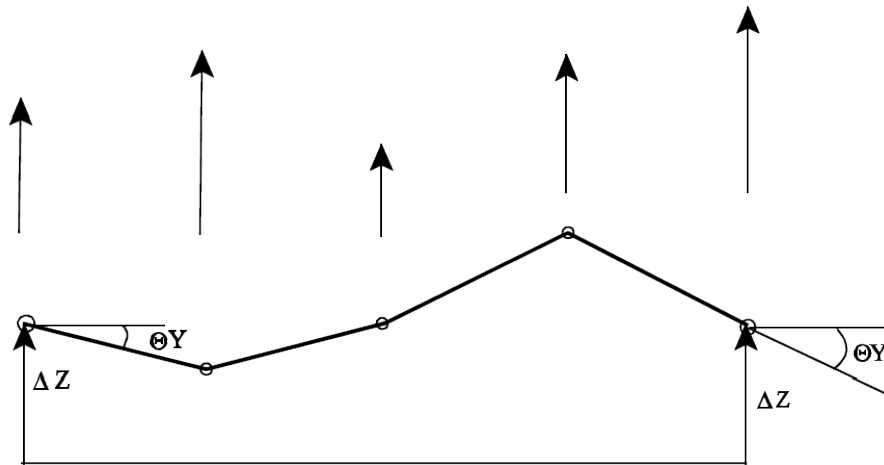
$$\begin{aligned}& \int (\sigma'_x \delta \epsilon_x + \sigma'_{xy} \delta \epsilon_{xy} + \sigma_x \delta \epsilon'_x + \sigma_{xy} \delta \epsilon'_{xy}) dVOL \\ & - P'_i \delta r_i = P_i \delta r_i + \int (\sigma_x \delta \epsilon_x + \sigma_{xy} \delta \epsilon_{xy}) dVOL\end{aligned}\quad (41)$$

Expressing stresses in terms of strains and utilizing the strain expressions in section 5.4.1 and integrating over the volume of the bar, equation (41) can be written in matrix form as:

$$[K_{inc}] \{\Delta q\} - \{\Delta P\} = \{P\} - [K_e] \{q\}\quad (42)$$

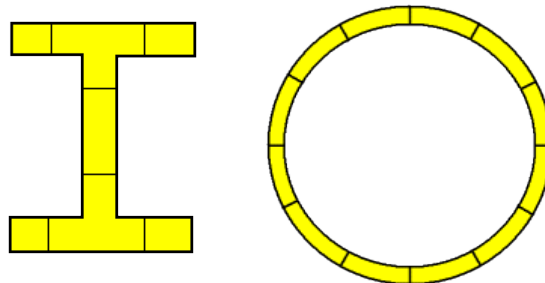
Equation (42) represents a set of linear simultaneous equations in the unknowns displacement increments $\{\Delta q\}$ and load increments $\{\Delta P\}$ which is composed of the applied load vector $\{P_a\}$ and the plastic load vector $\{P_p\}$. The left side of equation (42) represents the incremental equations of equilibrium and the right side represents the equilibrium equations which vanishes when the system is in a state of equilibrium. Equation (42) can be solved iteratively.

To account for the inter-nodal large displacement nonlinearities, the member element is divided along its length into sub-elements. The number of sub-elements is controlled by the user up to a maximum of 20, with a default of 8. This subdivision will allow the program to account for inter-nodal buckling as well as predict the contribution of the inter-nodal large displacements on the surrounding structure. Each member that is sub-divided essentially becomes a super-element to the structure. From the global stiffness analysis, the member end deflections and rotations are known as well as any inter-nodal loading.

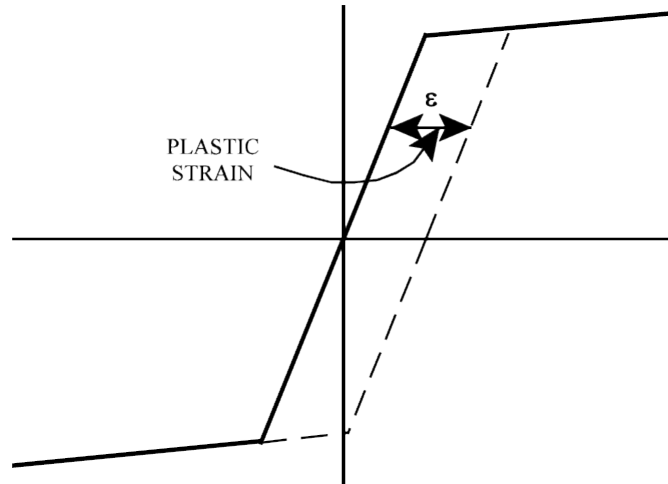


This super-element is solved iteratively using the end deflections and rotations and the intermediate loading until the internal deflections and rotations have converged. At each iteration, each sub-element is checked for plasticity as follows

- (a) The internal loads at each end of the sub-element is calculated.
- (b) The sub-element cross-section is divided into sub-areas and the axial and shear stress is calculated for each sub-area as shown below for wide flange and tubular cross-sections. Other cross-sections are similar.



(c) For each sub-area, the plasticity is determined by calculating the amount of strain which exceeds the von Mises-Hencky stress envelope. The plastic strain is retained for each subarea of each sub-element through-out the loading sequence to facilitate the unloading of a sub-area if required.



(d) If the local buckling is to be included, the strain is compared to the local buckling strain level of the following:^[11]

$$\epsilon_{lb} = \frac{16}{(D/t)^2} \quad (43)$$

If this value is exceeded, a hinge is formed and the sub-element will have zero moment capacity.

(e) The plastic stresses are then used to compute self-equilibrating plastic forces on each sub-element.

(f) These plastic forces are then used in the member iterative solution.

(g) After the final member iteration, the resulting end plastic forces are transformed into the global coordinates and added to the global plastic force vector.

(h) The final member stiffness is calculated on the final deflected position of the sub-elements.

4.5 CONNECTIONS

4.5.1 Joint Flexibility

The Collapse program can optionally consider the flexibility of a connection which has a tubular chord. The program utilizes equations for the connection flexibility as proposed by Fessler[12] which relate the local axial, in-plane moment, and out-of-plane moment to the corresponding deflection and rotation of the member ends. The following equations are used to calculate the local joint flexibility (LJF):

$$LJF_{\text{Axial}} = \frac{1.95\gamma^{2.15}(1-\beta)^{1.3}\sin^{2.19}\theta}{ED_C} \quad (44)$$

$$LJF_{\text{OPB}} = \frac{85.5\gamma^{2.2}\exp(-3.85\beta)\sin^{2.16}\theta}{ED_C^3} \quad (45)$$

$$LJF_{\text{IPB}} = \frac{134\gamma^{1.73}\exp(-4.52\beta)\sin^{1.22}\theta}{ED_C^3} \quad (46)$$

in which

$$\gamma = \frac{1}{2} \frac{D_C}{T_C} \quad \beta = \frac{D_B}{D_C} \quad (47)$$

where D_c and T_c are the chord diameter and thickness, respectively, D_b is the brace diameter, θ is the angle between the brace and chord and E is the chord elastic modulus.

Note: The flexibility of a connection with a non-tubular brace is determined using an equivalent brace diameter.

4.5.2 Tubular Connection Capacity

Collapse uses an ultimate limit state approach to check for tubular joint failure where chord and brace capacities are calculated based on either the API RP 2A-LRFD or the Norsok recommendations. For the API-LRFD standard, the connection capacity ratio is determined for the connection based on the following equation:^[13]

$$1 - \cos \left[\frac{\pi}{2} \left(\frac{P_D}{\phi_j P_{uj}} \right) \right] + \left[\left(\frac{M_D}{\phi_j P_{uj}} \right)_{\text{ipb}}^2 + \left(\frac{M_D}{\phi_j P_{uj}} \right)_{\text{opb}}^2 \right]^{\frac{1}{2}} > 1.0 \quad (48)$$

where the subscripts ipb and opb refer to in-plane bending and out-of-plane bending, respectively, P_D is the axial load in the brace member, P_{uj} is the ultimate joint axial capacity, M_D is the bending moment in the brace member, M_{uj} is the ultimate joint bending moment capacity and ϕ_j is the ultimate strength resistance factor for tubular joints. For the Norsok standard, the connection capacity ratio is determined for the connection based on the following inequality:^[14]

$$\frac{P_D}{\phi_j P_{uj}} + \left(\frac{M_D}{\phi_j P_{uj}} \right)_{\text{ipb}}^2 + \left(\frac{M_D}{\phi_j P_{uj}} \right)_{\text{opb}}^2 \leq 1.0 \quad (49)$$

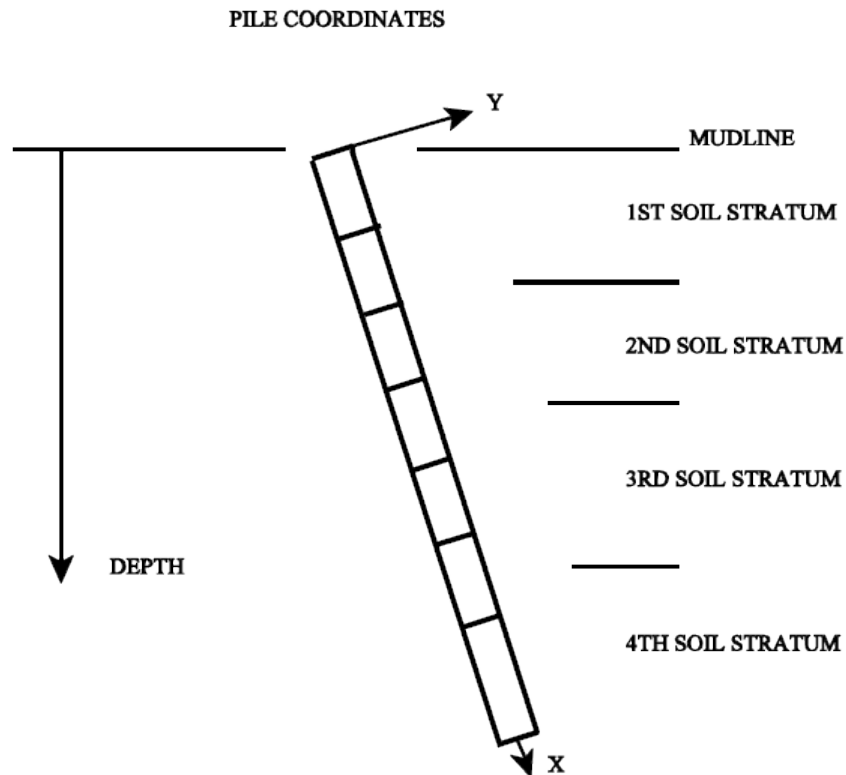
When the joint capacity ratio determined from equation (48) or (49) exceeds 1.0, the connection is considered to have failed. Once the connection has failed, the brace stiffness is removed from the analysis.

5 FOUNDATION

The effects of the nonlinear foundation including piles below the mud-line and the soil may be accounted for in the plastic collapse analysis.

5.1 PILE REPRESENTATION

The piles are represented structurally as segmented members using a full 3-D finite element approach including shear deformation as shown in the figure below:



The 3-D analysis allows the pile to deflect in any direction at any point down along the length of the pile.

5.2 SOIL REPRESENTATION

The axial soil representation can be either T-Z data where the soil resistance is a function of the axial displacement or adhesion data where the axial load in the pile is removed at the rate of the soil capacity. The T-Z approach would be preferred since the relative stiffness of the soil and the pile is represented. The end bearing is also represented by either a load versus deflection (T-Z) or as a total capacity. For the lateral soil data, the load versus deflection (P-Y) is used. Torsion of the pile is normally represented by a torsional spring.

6 SAMPLE PROBLEMS

This section presents some sample problems used to illustrate some of the features and capabilities of the Collapse program module. Two sample problems are detailed:

1. The first sample problem is a simply supported beam used to demonstrate the elasto-plastic behavior of the element.
2. The second sample problem is an environmental loading push over analysis of a frame type structure.

6.1 ELASTO-PLASTIC BEAM ANALYSIS

Sample Problem 1 illustrates an elasto-plastic beam analysis. This sample problem considers a simply supported beam with a point load at midspan. The beam is restrained in the axial direction so that membrane action is introduced at large deflections. The beam is of circular cross section and is modeled as two elements as shown in Figure 1 below.

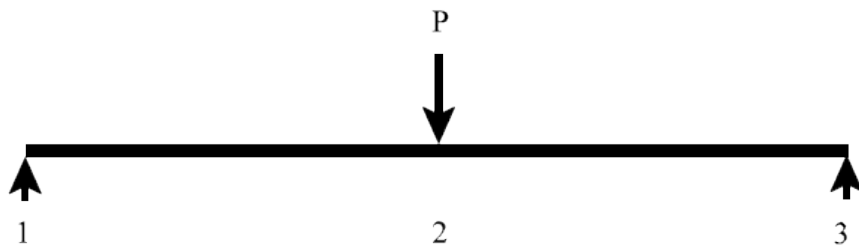


Figure 1

The Collapse model file for the simply supported beam follows:

```
1 1234567890123456789012345678901234567890123456789012345678901234567890
2 SIMPLY SUPPORTED BEAM SAMPLE
3 OPTIONS      MN    NL          1    DC    C
4 GRUP DIG          50.0000.500  21.007.72333.00 1      1.0 1.0      0.5N7.8490
5 MEMBER
6 MEMBER      1    2 DIG
7 MEMBER      2    3 DIG
8 JOINT
9 JOINT 1      0.    0.    0.          111101
10 JOINT 2     5.0    0.    0.          010101
11 JOINT 3     10.    0.    0.          111101
12 LOAD
13 LOADCN1
14 LOAD Z      2          -10.000          GLOB JOIN    PTLOAD
15 END
```

The Collapse input file containing the Collapse analysis input data is shown below:

```
1 1234567890123456789012345678901234567890123456789012345678901234567890
2 CLPOPT      80  8 20          ALSI          .002
3 CLPRPT P1R1M1
4 LDSEQ AAAA          1    250 0.00  90.0
5 END
```

The following is a description of selected input used in the Collapse input file for the sample problem:

Line 1. The collapse analysis options are specified on the line labeled CLPOPT as follows:

- a. The maximum number of iterations per load increment is set to 80 in columns 12-13 and the default number of member iterations is used (20 in columns 17-19).
- b. The number of segments per member is set to the default value of 8 and the default values for convergence criterion was used.
- c. Strain hardening ratio of .002 was specified in columns 76-80.

Line 2. The joint displacements, joint reactions and member stresses are reported at every load increment as designated by 'P1', 'M1' and 'R1' on the CLPRPT input line.

Line 3. The load sequence is input on the LDSEQ input line as follows:

- a. Load case 1 is to be applied in 250 increments starting with a load factor of 0.0 and ending with a load factor of 90.0
4. The end of input is designated by the input line labeled END.

Figure 2 below shows a color-coded plastic interaction plot of the sample problem generated by Collapse View, the interactive collapse view program.

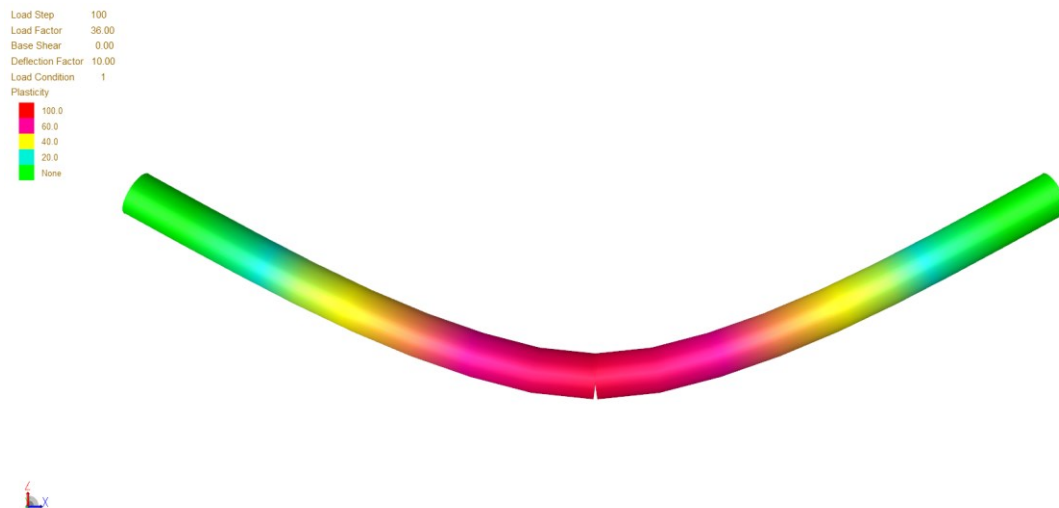


Figure 2

Figure 3 shows a typical load displacement plot generated by Collapse View, the Collapse interactive viewing program:

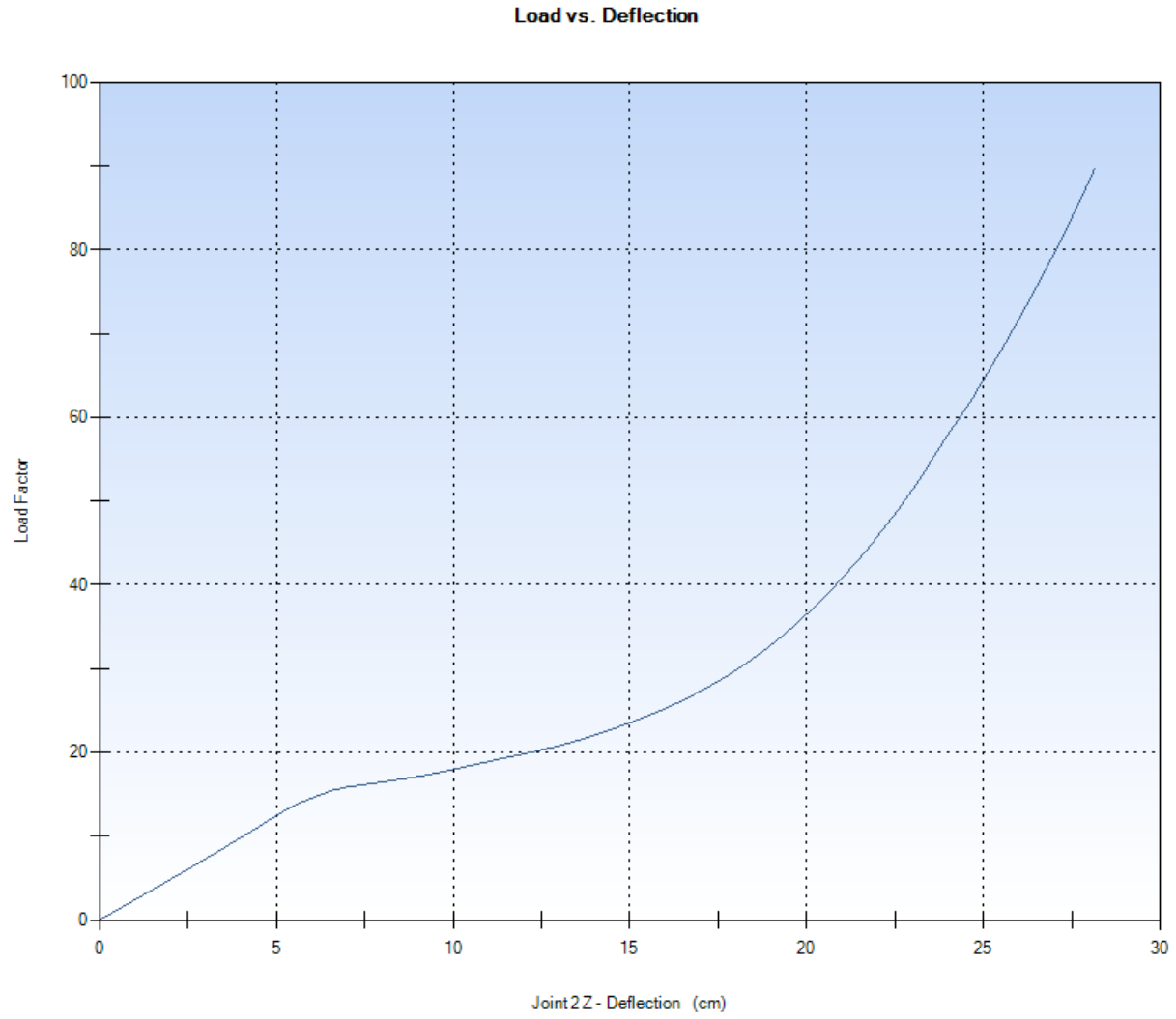


Figure 3

Portions from the Collapse output file follow:

** SACS COLLAPSE REACTION FORCES AND MOMENTS **																
			INCREMENT 34			LOAD FACTOR 12.240										
JOINT NO.	FORCE (X) KN		FORCE (Y) KN		FORCE (Z) KN		MOMENT (X) KN-M		MOMENT (Y) KN-M		MOMENT (Z) KN-M					
1	-188.049		0.000		61.200		0.000		0.000		0.000					
2	0.000		0.000		0.000		0.000		0.000		0.000					
3	188.049		0.000		61.200		0.000		0.000		0.000					
SACS CONNECT Edition V(14.2) - CL SIMPLY SUPPORTED BEAM SAMPLE										Company: DATE 14-APR-2020 TIME 13:59:32 CLP PAGE 138						
** SACS COLLAPSE MEMBER FORCES AND MOMENTS **																
			INCREMENT 34			LOAD FACTOR 12.240										
MEMBER	GRP	LOC. M	***** INTERNAL FORCES *****			***** INTERNAL MOMENTS *****			***** STRESSES *****							
			X KN	Y KN	Z KN	X KN-M	Y KN-M	Z KN-M	AXIAL N/MM2	BEND-Y N/MM2	BEND-Z N/MM2	SHEAR-Y N/MM2	SHEAR-Z N/MM2	PLAST. RATIO		
1-	2	DIG	0.00	189.152	0.000	58.453	0.0	0.2	0.0	24.3	0.2	0.0	0.0	15.0	0.00	
			0.62	189.152	0.000	58.453	0.0	36.7	0.0	24.3	38.5	0.0	0.0	15.0	0.00	
			1.25	189.125	0.000	58.540	0.0	73.3	0.0	24.3	76.9	0.0	0.0	15.1	0.00	
			1.87	189.071	0.000	58.713	0.0	110.0	0.0	24.3	115.4	0.0	0.0	15.1	0.00	
			2.50	188.990	0.000	58.973	0.0	146.8	0.0	24.3	154.1	0.0	0.0	15.2	0.00	
			3.12	188.882	0.000	59.320	0.0	183.9	0.0	24.3	193.1	0.0	0.0	15.3	0.00	
			3.75	188.745	0.000	59.755	0.0	221.3	0.0	24.3	232.3	0.0	0.0	15.4	0.00	
			4.37	188.579	0.000	60.277	0.0	258.9	0.0	24.3	271.8	0.0	0.0	15.5	0.00	
			5.00	188.330	0.000	60.880	0.0	297.0	0.0	24.2	311.7	0.0	0.0	15.7	0.08	
2-	3	DIG	0.00	187.025	0.000	-60.950	0.0	296.8	0.0	24.1	311.5	0.0	0.0	-15.7	0.08	
			0.62	187.025	0.000	-60.950	0.0	258.7	0.0	24.1	271.6	0.0	0.0	-15.7	0.08	
			1.25	188.830	0.000	-60.341	0.0	221.4	0.0	24.3	232.4	0.0	0.0	-15.5	0.00	
			1.87	188.997	0.000	-59.817	0.0	184.0	0.0	24.3	193.1	0.0	0.0	-15.4	0.00	
			2.50	189.134	0.000	-59.381	0.0	146.9	0.0	24.3	154.2	0.0	0.0	-15.3	0.00	
			3.12	189.243	0.000	-59.032	0.0	110.0	0.0	24.3	115.5	0.0	0.0	-15.2	0.00	
			3.75	189.324	0.000	-58.771	0.0	73.3	0.0	24.3	76.9	0.0	0.0	-15.1	0.00	
			4.37	189.378	0.000	-58.597	0.0	36.6	0.0	24.4	38.5	0.0	0.0	-15.1	0.00	
			5.00	189.405	0.000	-58.511	0.0	0.1	0.0	24.4	0.1	0.0	0.0	-15.1	0.00	
*** PLASTICITY OCCURRED ON MEMBER			1- 2 AT LOAD STEP			34										
*** PLASTICITY OCCURRED ON MEMBER			2- 3 AT LOAD STEP			34										

***** COLLAPSE SOLUTION SUMMARY *****															
LOAD SEQUENCE 1															
INCR	LOAD CASE	LOAD FACTOR	NO. LOOPS	* MAXIMUM DEFL. CM	DEFLECTION * JOINT DOF		** MAXIMUM ROT. JOINT DOF	ROTATION ** JOINT DOF		** SOLUTION DATA ** MAX. JOINT DOF DIGITS			*** REACTION SUMMATION *** FX KN FY KN FZ KN		
1	1	0.36	1	-0.150	2	DZ	0.0004499	1	RY	0	1	RY	0.00	0.00	3.60
2	1	0.72	1	-0.300	2	DZ	0.0008997	1	RY	0	1	RY	0.00	0.00	7.20
3	1	1.08	1	-0.450	2	DZ	0.0013494	1	RY	0	1	RY	0.00	0.00	10.80
4	1	1.44	1	-0.600	2	DZ	0.0017988	1	RY	0	1	RY	0.00	0.00	14.40
5	1	1.80	1	-0.749	2	DZ	0.0022480	1	RY	0	1	RY	0.00	0.00	18.00
6	1	2.16	1	-0.899	2	DZ	0.0026967	1	RY	0	1	RY	0.00	0.00	21.60
7	1	2.52	1	-1.048	2	DZ	0.0031449	1	RY	0	1	RY	0.00	0.00	25.20
8	1	2.88	1	-1.198	2	DZ	0.0035924	1	RY	0	1	RY	0.00	0.00	28.80
9	1	3.24	1	-1.347	2	DZ	0.0040393	1	RY	0	1	RY	0.00	0.00	32.40
10	1	3.60	1	-1.495	2	DZ	0.0044853	1	RY	0	1	RY	0.00	0.00	36.00
11	1	3.96	1	-1.644	2	DZ	0.0049305	1	RY	0	1	RY	0.00	0.00	39.60
12	1	4.32	1	-1.792	2	DZ	0.0053747	1	RY	0	1	RY	0.00	0.00	43.20
13	1	4.68	1	-1.940	2	DZ	0.0058178	1	RY	0	1	RY	0.00	0.00	46.80
14	1	5.04	1	-2.087	2	DZ	0.0062597	1	RY	0	1	RY	0.00	0.00	50.40
15	1	5.40	1	-2.234	2	DZ	0.0067005	1	RY	0	1	RY	0.00	0.00	54.00
16	1	5.76	1	-2.381	2	DZ	0.0071398	1	RY	0	1	RY	0.00	0.00	57.60
17	1	6.12	1	-2.527	2	DZ	0.0075778	1	RY	0	1	RY	0.00	0.00	61.20
18	1	6.48	1	-2.673	2	DZ	0.0080144	1	RY	0	1	RY	0.00	0.00	64.80
19	1	6.84	1	-2.818	2	DZ	0.0084493	1	RY	0	1	RY	0.00	0.00	68.40
20	1	7.20	1	-2.962	2	DZ	0.0088827	1	RY	0	1	RY	0.00	0.00	72.00
21	1	7.56	1	-3.107	2	DZ	0.0093143	1	RY	0	1	RY	0.00	0.00	75.60
22	1	7.92	1	-3.250	2	DZ	0.0097442	1	RY	0	1	RY	0.00	0.00	79.20
23	1	8.28	1	-3.393	2	DZ	0.0101723	1	RY	0	1	RY	0.00	0.00	82.80
24	1	8.64	1	-3.536	2	DZ	0.0105985	1	RY	0	1	RY	0.00	0.00	86.40
25	1	9.00	1	-3.677	2	DZ	0.0110228	1	RY	0	1	RY	0.00	0.00	90.00
26	1	9.36	1	-3.818	2	DZ	0.0114451	1	RY	0	1	RY	0.00	0.00	93.60
27	1	9.72	1	-3.959	2	DZ	0.0118654	1	RY	0	1	RY	0.00	0.00	97.20
28	1	10.08	1	-4.099	2	DZ	0.0122836	1	RY	0	1	RY	0.00	0.00	100.80
29	1	10.44	1	-4.238	2	DZ	0.0126997	1	RY	0	1	RY	0.00	0.00	104.40
30	1	10.80	1	-4.376	2	DZ	0.0131135	1	RY	0	1	RY	0.00	0.00	108.00
31	1	11.16	1	-4.514	2	DZ	0.0135252	1	RY	0	1	RY	0.00	0.00	111.60
32	1	11.52	1	-4.651	2	DZ	0.0139347	1	RY	0	1	RY	0.00	0.00	115.20
33	1	11.88	1	-4.787	2	DZ	0.0143418	1	RY	0	1	RY	0.00	0.00	118.80
34	1	12.24	1	-4.923	2	DZ	0.0147467	1	RY	0	1	RY	0.00	0.00	122.40
35	1	12.60	1	-5.061	2	DZ	0.0151570	1	RY	0	1	RY	0.00	0.00	126.00
36	1	12.96	1	-5.207	2	DZ	0.0155831	1	RY	0	1	RY	0.00	0.00	129.60
37	1	13.32	1	-5.353	2	DZ	0.0160062	1	RY	0	1	RY	0.00	0.00	133.20
38	1	13.68	2	-5.523	2	DZ	-0.0164725	3	RY	0	1	RY	0.00	0.00	136.80
39	1	14.04	2	-5.715	2	DZ	-0.0169947	3	RY	0	1	RY	0.00	0.00	140.40
40	1	14.40	2	-5.925	2	DZ	-0.0175533	3	RY	0	1	RY	0.00	0.00	144.00

6.2 ELASTO-PLASTIC JACKET ANALYSIS

This sample problem illustrates an elasto-plastic jacket push-over analysis. It considers a simple jacket structure shown below. The jacket is initially loaded with self-weight, area, equipment, live, and miscellaneous loads. A horizontal wave load acting on the structure is then incremented until collapse occurs.

Load Condition: S000

Wave: STRE

Direction 0.000 (deg)

Height 50.000 (ft)

Period 13.000 (sec)

Water depth 266.000 (ft)

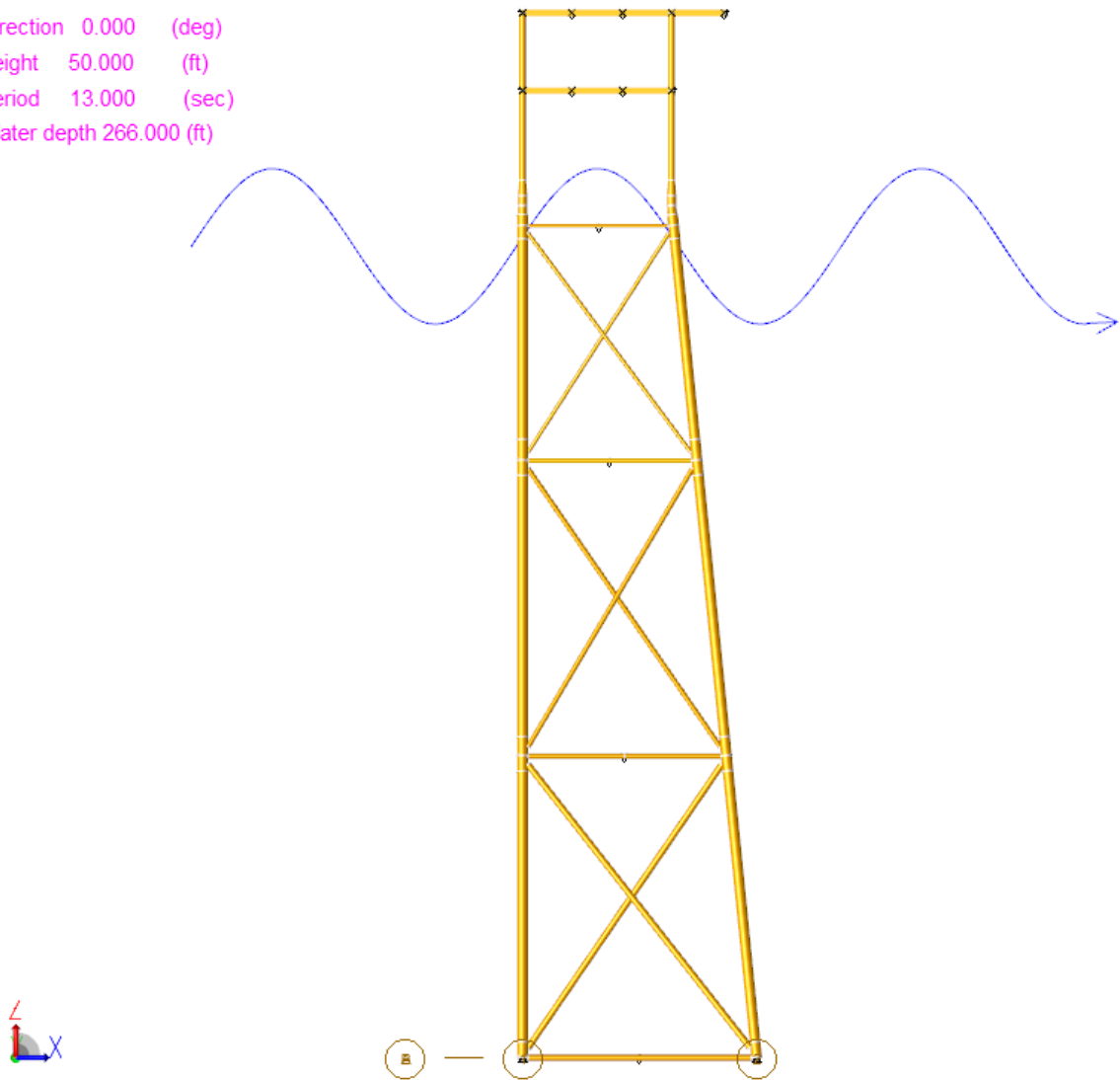


Figure 4

The Collapse load data from the model file for the jacket is shown below.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	LOADCNAREA							

3	*							
4	***LDS1**	24.000	-26.247	50.000	24.000	26.247	50.000	-24.000
5	***LDS2**	-26.247	50.000	-24.000	26.247	50.000	-10.000	
6	***LDS3**		0 1 3 0	0AREA	-2EQUPPRES10PSFL			
7	LOAD Z 701 705		-0.0790	-0.0790		GLOB UNIF	10PSFL	
8	LOAD Z 703 707		-0.0790	-0.0790		GLOB UNIF	10PSFL	
9	LOAD Z 705 720		-0.0790	-0.0790		GLOB UNIF	10PSFL	
10	LOAD Z 707 723		-0.0790	-0.0790		GLOB UNIF	10PSFL	
11	LOAD Z 709 701		-0.0790	-0.0790		GLOB UNIF	10PSFL	
12	LOAD Z 710 714		-0.1610	-0.1610		GLOB UNIF	10PSFL	
13	LOAD Z 711 715		-0.1610	-0.1610		GLOB UNIF	10PSFL	
14	LOAD Z 712 703		-0.0790	-0.0790		GLOB UNIF	10PSFL	
15	LOAD Z 714 717		-0.1610	-0.1610		GLOB UNIF	10PSFL	
16	LOAD Z 715 718		-0.1610	-0.1610		GLOB UNIF	10PSFL	
17	LOAD Z 717 721		-0.1610	-0.1610		GLOB UNIF	10PSFL	
18	LOAD Z 718 722		-0.1610	-0.1610		GLOB UNIF	10PSFL	
19	*							
20	***LDS1**	41.011	-26.247	75.000	41.011	26.247	75.000	-24.000
21	***LDS2**	-26.247	75.000	-24.000	26.247	75.000	-15.000	
22	***LDS3**		0 1 3 0	0AREA	-2EQUPPRES15PSFU			
23	LOAD Z 801 805		-0.1180	-0.1180		GLOB UNIF	15PSFU	
24	LOAD Z 803 807		-0.2460	-0.2460		GLOB UNIF	15PSFU	
25	LOAD Z 805 840		-0.1180	-0.1180		GLOB UNIF	15PSFU	
26	LOAD Z 807 843		-0.2460	-0.2460		GLOB UNIF	15PSFU	
27	LOAD Z 829 801		-0.1180	-0.1180		GLOB UNIF	15PSFU	
28	LOAD Z 830 834		-0.2420	-0.2420		GLOB UNIF	15PSFU	
29	LOAD Z 831 835		-0.2420	-0.2420		GLOB UNIF	15PSFU	
30	LOAD Z 832 803		-0.2460	-0.2460		GLOB UNIF	15PSFU	
31	LOAD Z 833 836		-0.1280	-0.1280		GLOB UNIF	15PSFU	
32	LOAD Z 834 837		-0.2420	-0.2420		GLOB UNIF	15PSFU	
33	LOAD Z 835 838		-0.2420	-0.2420		GLOB UNIF	15PSFU	
34	LOAD Z 836 839		-0.1280	-0.1280		GLOB UNIF	15PSFU	
35	LOAD Z 837 841		-0.2420	-0.2420		GLOB UNIF	15PSFU	
36	LOAD Z 838 842		-0.2420	-0.2420		GLOB UNIF	15PSFU	
37	LOAD Z 839 844		-0.1280	-0.1280		GLOB UNIF	15PSFU	
38	LOADCNEQPT							
39	*							
40	***LDS1**	16.000	6.000	75.000	16.000	6.000	75.000	
41	***LDS2**		-250.000				20.000	10.000
42	***LDS3**	10.000	1 2 2 0	0EQPT	-1EQUPSKIDSKID1	X		
43	LOAD Z 835 838	17.4040-65.579				GLOB CONC	SKID1	
44	LOAD Z 835 838	27.4040-65.579				GLOB CONC	SKID1	
45	LOAD Z 803 807	17.4040-59.421				GLOB CONC	SKID1	
46	LOAD Z 803 807	27.4040-59.421				GLOB CONC	SKID1	
47	*							
48	***LDS1**	-16.000	-16.000	75.000	-16.000	-16.000	75.000	
49	***LDS2**		-150.000				20.000	7.500
50	***LDS3**	7.500	1 2 2 0	0EQPT	-1EQUPSKIDSKID2	X		
51	LOAD Z 829 801	6.49700-35.653				GLOB CONC	SKID2	
52	LOAD Z 830 834	6.49700-39.347				GLOB CONC	SKID2	
53	LOAD Z 834 837	4.15400-39.347				GLOB CONC	SKID2	
54	LOAD Z 801 805	4.15400-35.653				GLOB CONC	SKID2	
55	*							
56	***LDS1**	-16.000		50.000	-16.000		50.000	
57	***LDS2**		-100.000				20.000	7.500
58	***LDS3**	7.500	1 2 2 0	0EQPT	-1EQUPSKIDSKID3	X		
59	LOAD Z 701 705	12.6540-23.769				GLOB CONC	SKID3	
60	LOAD Z 701 705	20.1540-23.769				GLOB CONC	SKID3	
61	LOAD Z 714 717	12.6540-26.231				GLOB CONC	SKID3	
62	LOAD Z 714 717	20.1540-26.231				GLOB CONC	SKID3	
63	*							
64	***LDS1**	32.000	19.000	75.000	32.000	19.000	75.000	
65	***LDS2**		-35.000				20.000	7.000
66	***LDS3**	7.000	1 2 3 0	0EQPT	-1EQUPSKIDSKID4	X		
67	LOAD Z 803 807	31.5000-6.1800				GLOB CONC	SKID4	
68	LOAD Z 807 843	3.00000-6.1800				GLOB CONC	SKID4	
69	LOAD Z 807 843	6.50000-6.1800				GLOB CONC	SKID4	
70	LOAD Z 836 839	31.5000-5.4866				GLOB CONC	SKID4	
71	LOAD Z 839 844	3.00000-5.4866				GLOB CONC	SKID4	
72	LOAD Z 839 844	6.50000-5.4866				GLOB CONC	SKID4	

73	LOADCNLIVE							
74	*							
75	***LDS1**	41.011	-26.247	75.000	41.011	26.247	75.000	-24.606
76	***LDS2**	26.247	75.000	-24.606	-26.247	75.000	-100.000	
77	***LDS3**		0 1 3 0	0LIVE	-2EQUPPRES	100PSFU		
78	LOAD Z 829 801		-0.8200	-0.8200		GLOB UNIF	100PSFU	
79	LOAD Z 830 834		-1.6400	-1.6400		GLOB UNIF	100PSFU	
80	LOAD Z 831 835		-1.6400	-1.6400		GLOB UNIF	100PSFU	
81	LOAD Z 832 803		-1.6400	-1.6400		GLOB UNIF	100PSFU	
82	LOAD Z 833 836		-0.8200	-0.8200		GLOB UNIF	100PSFU	
83	LOAD Z 834 837		-1.6400	-1.6400		GLOB UNIF	100PSFU	
84	LOAD Z 835 838		-1.6400	-1.6400		GLOB UNIF	100PSFU	
85	LOAD Z 836 839		-0.8200	-0.8200		GLOB UNIF	100PSFU	
86	LOAD Z 837 841		-1.6400	-1.6400		GLOB UNIF	100PSFU	
87	LOAD Z 838 842		-1.6400	-1.6400		GLOB UNIF	100PSFU	
88	LOAD Z 839 844		-0.8200	-0.8200		GLOB UNIF	100PSFU	
89	LOAD Z 801 805		-0.8200	-0.8200		GLOB UNIF	100PSFU	
90	LOAD Z 803 807		-1.6400	-1.6400		GLOB UNIF	100PSFU	
91	LOAD Z 805 840		-0.8200	-0.8200		GLOB UNIF	100PSFU	
92	LOAD Z 807 843		-1.6400	-1.6400		GLOB UNIF	100PSFU	
93	*							
94	***LDS1**	24.606	-26.247	50.000	24.606	26.247	50.000	-24.606
95	***LDS2**	26.247	50.000	-24.606	-26.247	50.000	-50.000	
96	***LDS3**		0 1 3 0	0LIVE	-2EQUPPRES	50PSFL		
97	LOAD Z 701 705		-0.4100	-0.4100		GLOB UNIF	50PSFL	
98	LOAD Z 703 707		-0.4100	-0.4100		GLOB UNIF	50PSFL	
99	LOAD Z 705 720		-0.4100	-0.4100		GLOB UNIF	50PSFL	
100	LOAD Z 707 723		-0.4100	-0.4100		GLOB UNIF	50PSFL	
101	LOAD Z 709 701		-0.4100	-0.4100		GLOB UNIF	50PSFL	
102	LOAD Z 710 714		-0.8200	-0.8200		GLOB UNIF	50PSFL	
103	LOAD Z 711 715		-0.8200	-0.8200		GLOB UNIF	50PSFL	
104	LOAD Z 712 703		-0.4100	-0.4100		GLOB UNIF	50PSFL	
105	LOAD Z 714 717		-0.8200	-0.8200		GLOB UNIF	50PSFL	
106	LOAD Z 715 718		-0.8200	-0.8200		GLOB UNIF	50PSFL	
107	LOAD Z 717 721		-0.8200	-0.8200		GLOB UNIF	50PSFL	
108	LOAD Z 718 722		-0.8200	-0.8200		GLOB UNIF	50PSFL	
109	LOADCNMISC							
110	LOAD Z 712 703		-0.1900	-0.1900		GLOB UNIF	WALK1	
111	LOAD Z 703 707		-0.1900	-0.1900		GLOB UNIF	WALK1	
112	LOAD Z 707 723		-0.1900	-0.1900		GLOB UNIF	WALK1	
113	LOAD Z 833 836		-0.1900	-0.1900		GLOB UNIF	WALK2	
114	LOAD Z 836 839		-0.1900	-0.1900		GLOB UNIF	WALK2	
115	LOAD Z 839 844		-0.1900	-0.1900		GLOB UNIF	WALK2	
116	LOAD 807			-20.000		GLOB JOIN	CRANE	
117	*							
118	***LDS1**	-8.000	20.000	50.000	-8.000	20.000	50.000	
119	***LDS2**		-10.000				34.000	0.100
120	***LDS3**	0.100	1 2 2 0	0MISC	-1EQUPSKID	FIREWALLX		
121	LOAD Z 705 720	3.95000	-1.6667			GLOB CONC	FIREWALL	
122	LOAD Z 705 720	4.05000	-1.6667			GLOB CONC	FIREWALL	
123	LOAD Z 717 721	3.95000	-1.6667			GLOB CONC	FIREWALL	
124	LOAD Z 717 721	4.05000	-1.6667			GLOB CONC	FIREWALL	
125	LOAD Z 718 722	3.95000	-1.6667			GLOB CONC	FIREWALL	
126	LOAD Z 718 722	4.05000	-1.6667			GLOB CONC	FIREWALL	

The following is a description of selected input lines used in the load data:

Line 2. Load condition AREA represents the application of general area loads to both decks.

Line 38. Load condition EQPT represent the application of a equipment loads to the decks.

Line 73. Load condition LIVE represents the application of live loads to both decks.

Line 109. Load condition MISC represents the application of miscellaneous dead loads like the crane, a firewall and handrails.

The Collapse load data from the seastate input file for the jacket is shown below.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	LDOPT	NF+Z64.30001490.0500-261.000	261.000	GLOBEN				
3	* USE LOADING IN SEASTATE INPUT AND MODEL FILE							
4	FILE B							
5	CDM							
6	CDM	1.00 0.600	1.200	0.600	1.200			
7	MGROV							
8	MGROV	0.000	200.000	1.000				
9	MGROV	200.000	261.000	2.000				
10	GRPOV							
11	GRPOVAL	LG1 F						
12	GRPOVAL	LG2 F						
13	GRPOVAL	LG3 F						
14	GRPOV	LG4 F						
15	GRPOV	PL1NF		0.001	0.001			
16	GRPOV	PL2NF		0.001	0.001			
17	GRPOV	PL3NF		0.001	0.001			
18	GRPOV	PL4NF		0.001	0.001			
19	GRPOV	W.BNF 0.001		0.001	0.001			
20	LOAD							
21	LOADCNDEAD							
22	DEAD							
23	DEAD	-Z		M				
24	LOADCNS000							
25	WAVE							
26	WAVE	STRE 50.00266.00 13.00			L -75.00	5.00	20MS10 1	7
27	END							
28	END							

The following is a description of selected input lines used in the load data:

Line 20. Load condition DEAD represents the application of the structure self-weight and buoyancy.

Line 24. Load condition S000 represents the application of the wave load to the structure.

The collapse input file containing the collapse analysis data is shown below.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	CLPOPT	20 8 20	JF JS					100. .005
3	CLPRPT	MOMP SM						
4	* CREATE SEQUENCE THAT ADDS LOADING TO STRUCTURE							
5	LDSEQ AAAA	DEAD 1	1.0	EQPT 1	1.0	MISC 1	1.0	
6	LDSEQ	AREA 1	1.0	LIVE 1	0.5	S000 75	7.5	
7	* DECK WIDE FLANGES REMAIN ELASTIC							
8	GRELA	W01 W02 LG7						
9	* PLATES REMAIN ELASTIC							
10	PGRELA	P01						
11	END							

The following is a description of input lines used in the Collapse input file:

Line 1. Collapse analysis options were designated on the CLPOPT line as follows:

- The maximum number of iterations per load increment is set to 20 while the maximum number of member iterations is set to the default value of 20.
- The number of segments per member is set to the default value of 8.
- The effects of joint flexibility and joint strength are to be considered with 'JF' and 'JS' respectively.

d. The default values for deflection tolerance, rotation tolerance and member deflection tolerance are to be used.

e. The maximum deflection allowed before the structure is considered collapse is set to 100. In columns 71-75.

f. The strain hardening ratio is set to 0.005 in columns 76-80.

Line 2. Output reports consisting of member and plate final stresses 'MO', limited to plastic members 'MP', and a collapse summary report 'SM' on the CLPRPT line.

Line 4. The load sequence to be analyzed, named AAAA, is defined on the first LDSEQ input line as follows:

a. Load case DEAD is to be applied in 1 increment starting with a load factor of 0.0 and ending with a load factor of 1.0.

b. Load case EQPT is to be applied in 1 increment starting with a load factor of 0.0 and ending with a load factor of 1.0.

c. Load case MISC is to be applied in 1 increment starting with a load factor of 0.0 and ending with a load factor of 1.0.

Line 5. Load sequence AAAA is continued on the second LDSEQ input line.

a. Load case AREA is to be applied in 1 increment starting with a load factor of 0.0 and ending with a load factor of 1.0.

a. Load case LIVE is to be applied in 1 increment starting with a load factor of 0.0 and ending with a load factor of 0.5.

c. Finally the wave load, load case S000, is applied over 75 increments up to a maximum load factor of 7.5 as the final load step.

Line 7. The GRPELA (or GRPDEL) input line specifies that the material for all beam elements assigned to groups W01, W02, and LG7 are to remain elastic throughout the analysis. These are the deck elements which are not critical for a pushover analysis.

Line 9. The PGRELA input line specifies the material for all plates assigned to plate group P01 are to remain elastic throughout the analysis.

Line 10. the END line designates the end of input data.

Figure 5 shows color coded plastic interaction plots for load increment 50.

Note: The nonlinear plastic analysis results may be viewed in a 3D interactive graphical environment using the Collapse View program.

Load Step 50
Load Factor 4.50
Base Shear 1002.36
Deflection Factor 2.00
Load Condition S000

Plasticity
90.0
75.0
50.0
25.0
None

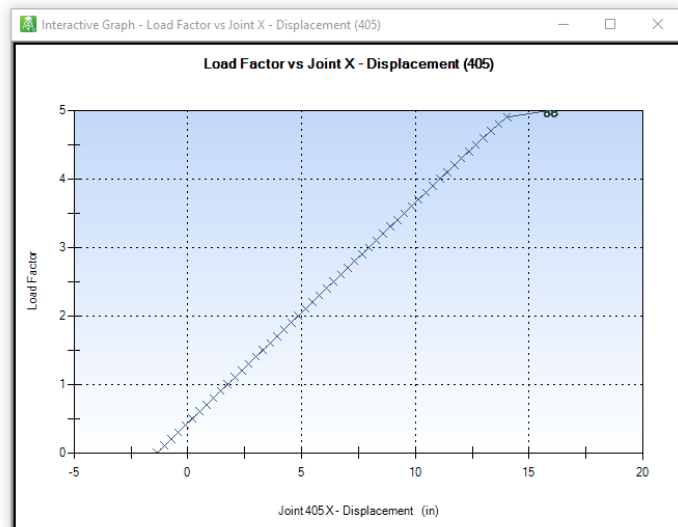
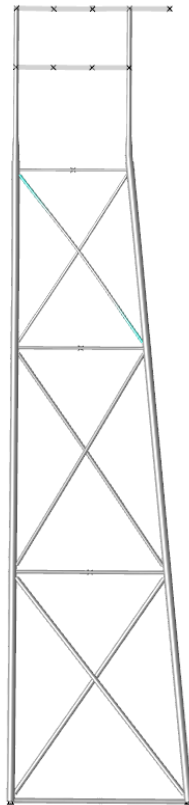


Figure 5

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SAMPLE PROBLEMS

8 INPUT LINES

JOINT FLEXIBILITY BRACE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE ALLOWS THE USER TO INCLUDE OR EXCLUDE BRACES WHEN CONSIDERING JOINT FLEXIBILITY. ONLY THE BRACE AND BRACE SIDE SPECIFIED ARE INCLUDED OR EXCLUDED PROVIDED ONE OF THE JOINT FLEXIBILITY OPTIONS IS SPECIFIED ON THE CLPOPT LINE. THIS LINE IS IGNORED IF NO FLEXIBILITY OPTION IS SPECIFIED.
(1- 5)	ENTER 'BFSEL'.
(7)	ENTER 'I' TO INCLUDE OR 'X' TO EXCLUDE THE BRACES SPECIFIED.
NOTE:	THE INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND CANNOT BE USED TOGETHER. ALL BRACES SPECIFIED ON BFSEL LINES MUST BE EITHER INCLUDED OR EXCLUDED.
(9-12)	ENTER THE BEGIN JOINT OF THE FIRST BRACE TO INCLUDE OR EXCLUDE.
(13-16)	ENTER THE END JOINT OF THE FIRST BRACE TO INCLUDE OR EXCLUDE.
(17-20)	ENTER THE JOINT DESIGNATING THE BRACE CONNECTION TO INCLUDE OR EXCLUDE. ENTER THE BEGIN JOINT TO CONSIDER THE CONNECTION AT THE START OF THE BRACE OR THE END JOINT TO CONSIDER THE CONNECTION AT THE END OF THE BRACE.
(22-72)	ENTER ADDITIONAL BRACES TO BE INCLUDED OR EXCLUDED.

LINE LABEL	INCLUDE OR EXCLUDE	1ST BRACE			2ND BRACE			3RD BRACE			4TH BRACE			5TH BRACE		
		BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION
BFSEL																
1-- 5	7	9-->12	13-->16	17-->20	22-->25	26-->29	30-->33	35-->38	39-->42	43-->46	48-->51	52-->55	56-->59	61-->64	65-->68	69-->72

JOINT STRENGTH BRACE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE ALLOWS THE USER TO INCLUDE OR EXCLUDE BRACES WHEN CONSIDERING JOINT STRENGTH. ONLY THE BRACE AND BRACE SIDE SPECIFIED ARE INCLUDED OR EXCLUDED PROVIDED ONE OF THE JOINT STRENGTH OPTIONS IS SPECIFIED ON THE CLPOPT LINE. THIS LINE IS IGNORED IF NO STRENGTH OPTION IS SPECIFIED.
(1- 5)	ENTER 'BSSEL'.
(7)	ENTER 'I' TO INCLUDE OR 'X' TO EXCLUDE THE BRACES SPECIFIED.
NOTE:	THE INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND CANNOT BE USED TOGETHER. ALL BRACES SPECIFIED ON BSSEL LINES MUST BE EITHER INCLUDED OR EXCLUDED.
(9-12)	ENTER THE BEGIN JOINT OF THE FIRST BRACE TO INCLUDE OR EXCLUDE.
(13-16)	ENTER THE END JOINT OF THE FIRST BRACE TO INCLUDE OR EXCLUDE.
(17-20)	ENTER THE JOINT DESIGNATING THE BRACE CONNECTION TO INCLUDE OR EXCLUDE. ENTER THE BEGIN JOINT TO CONSIDER THE CONNECTION AT THE START OF THE BRACE OR THE END JOINT TO CONSIDER THE CONNECTION AT THE END OF THE BRACE.
(22-72)	ENTER ADDITIONAL BRACES TO BE INCLUDED OR EXCLUDED.

LINE LABEL	INCLUDE OR EXCLUDE	1ST BRACE			2ND BRACE			3RD BRACE			4TH BRACE			5TH BRACE		
		BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION	BEGIN JOINT	END JOINT	CONNECTION
BSSEL																
1-- 5	7	9-->12	13-->16	17-->20	22-->25	26-->29	30-->33	35-->38	39-->42	43-->46	48-->51	52-->55	56-->59	61-->64	65-->68	69-->72

COLLAPSE ANALYSIS INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY ADDITIONAL OVERALL ANALYSIS PARAMETERS.
(8-13)	ENTER THE MEMBER ECCENTRICITY RATIO WHICH IS DEFINED AS: ER = E * C / R**2 WHERE E - ECCENTRICITY C - DISTANCE FROM NEUTRAL AXIS TO EXTREME FIBER R - RADIUS OF GYRATION
(15-20)	ENTER MAXIMUM DUCTILITY ALLOWED FOR ANY MEMBER. ANY MEMBER THAT EXCEEDS THIS LIMIT WILL BE TREATED AS FRACTURED.
(22-24)	ENTER "ITC" TO INCLUDE HIGHER ORDER CO-ROTATIONAL TERMS FOR TUBULAR AND CLOSED SECTIONS IN ENHANCED COLLAPSE. NOTE THAT IT IS ALREADY INCLUDED FOR OTHER SECTIONS.
(26-28)	ENTER "NWT" TO EXCLUDE WAGNER NONLINEAR TORSION STRAIN FROM THIN-WALLED OPEN SECTIONS IN ENHANCED COLLAPSE.

LINE LABEL	MEMBER ECCENTRICITY RATIO	MAXIMUM DUCTILITY ALLOWED	LEAVE BLANK
CLPOP2			
1-- 6	8<--13	15<--20	22-----80
DEFAULT			
ENGLISH		PERCENT	
METRIC	0.25	PERCENT	

COLLAPSE ANALYSIS INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY THE OVERALL ANALYSIS PARAMETERS.
(11-13)	ENTER MAXIMUM NUMBER OF ITERATIONS FOR EACH LOADING INCREMENT.
(14-16)	ENTER THE MINIMUM NUMBER OF SEGMENTS THAT A MEMBER IS TO BE DIVIDED FOR THE NONLINEAR MEMBER ANALYSIS. A NON-PRISMATIC MEMBER MAY RESULT IN MORE THAN THIS NUMBER SINCE EACH SEGMENT WILL BE LESS THAN OR EQUAL TO THE SECTION LENGTH.
(17-19)	ENTER THE MAXIMUM NUMBER OF ITERATIVE LOOPS FOR EACH MEMBER ANALYSIS.
(26-41)	SELECT FROM THE FOLLOWING ANALYSIS OPTIONS: 'LB' - LOCAL BUCKLING EFFECTS INCLUDED. 'JF' - JOINT FLEXIBILITY EFFECTS INCLUDED. 'FF' - JOINT FLEXIBILITY EFFECTS FROM SINGLE BRACE FORMULATION DUE TO FESSLER, MOCKFORD AND WEBSTER. 'BF' - JOINT FLEXIBILITY EFFECTS FROM SINGLE BRACE FORMULATION DUE TO BUITRAGO, HEALY AND CHANG. 'NS' - SKIPPED MEMBERS NOT TREATED AS LINEAR. 'PP' - INCLUDE PILE PLASTICITY. 'CN' - CONTINUE IF MAXIMUM NUMBER OF ITERATIONS IS EXCEEDED. 'JS' - JOINT STRENGTH CHECK (API-LRFD). 'N1' - JOINT STRENGTH CHECK (NORSOK N-004 REV. 1). 'ND' - JOINT STRENGTH CHECK (NORSOK N-004 REV. 2). 'N3' - JOINT STRENGTH CHECK (NORSOK N-004 REV. 3). 'IS' - JOINT STRENGTH CHECK (ISO 19902) 'DY' - DYNAMIC ANALYSIS OPTION. 'ME' - ALL MEMBERS ELASTIC. 'PE' - ALL PLATES ELASTIC. 'EB' - ELASTIC BUCKLING MONITOR

COLUMNS	COMMENTARY
(50-51)	ENTER 'SF' TO CREATE SACS IV FILE WITH FINAL DEFLECTED SHAPE.
(52-53)	IF THE LOCAL BUCKLING OPTION 'LB' IS REQUESTED, ENTER THE METHOD USED TO DETERMINE LOCAL BUCKLING CRITERIA AS: 'MG' - MARSHALL GATES LOWER LIMIT OF CRITICAL STRAIN. '2U' - API BULLETIN 2U. 'LR' - API LRFD. 'IS' - ISO 19902 SECTION 13.2.3.3.
(56-60)	ENTER THE DEFLECTION TOLERANCE REQUIRED FOR CONVERGENCE OF ANY LOAD INCREMENT.
(61-65)	ENTER THE ROTATION TOLERANCE REQUIRED FOR CONVERGENCE OF ANY LOAD INCREMENT.
(66-70)	ENTER THE CONVERGENCE TOLERANCE FOR THE MEMBERS.
(71-75)	ENTER THE MAXIMUM DEFLECTION ALLOWED BEFORE THE STRUCTURE IS CONSIDERED COLLAPSED.
(76-80)	ENTER THE STRAIN HARDENING RATIO. THIS IS THE RATIO OF THE SLOPE OF THE PLASTIC PORTION OF THE STRESS-STRAIN CURVE TO THE SLOPE OF THE ELASTIC PORTION.

LINE LABEL	ANALYSIS PARAMETERS			ANALYSIS OPTIONS								CREATE MODEL FILE	LOCAL BUCKLING METHOD	CONVERGENCE CRITERIA			COLLAPSE DEFLECTION	STRAIN HARDENING RATIO
	MAXIMUM NUMBER OF ITERATIONS PER LOAD INCREMENT	NUMBER OF MEMBER SEGMENTS	MAXIMUM NUMBER OF MEMBER ITERATIONS	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH			DEFLECTION TOLERANCE	ROTATION TOLERANCE	MEMBER DEFLECTION TOLERANCE		
CLPOPT																		
1-- 6	11-->13	14-->16	17-->19	26--27	28--29	30--31	32--33	34--35	36--37	38--39	40--41	50--51	52--53	56<--60	61<--65	66<--70	71<--75	76<--80
DEFAULT	20	8	20											0.01 ENG	0.001	0.01 ENG	1000.0 ENG	
ENGLISH														IN	RAD	IN	IN	
METRIC														CM	RAD	CM	CM	

COLLAPSE ANALYSIS REPORT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY THE COLLAPSE OUTPUT REPORT SELECTIONS.
(8-29)	SELECT FROM THE FOLLOWING OUTPUT REPORT CHOICES: JOINT DISPLACEMENT REPORT OPTION: 'P0' - PRINT FINAL DEFLECTIONS ONLY (DEFAULT) 'P1' - PRINT EVERY LOAD INCREMENT 'P2' - PRINT EVERY LOOP AND EVERY LOAD INCREMENT JOINT REACTION REPORT OPTION: 'R0' - PRINT FINAL REACTIONS (DEFAULT) 'R1' - PRINT REACTIONS AT EACH LOAD INCREMENT 'R2' - PRINT REACTIONS AT EACH LOOP MEMBER STRESSES AND INTERNAL LOADS REPORT OPTION: 'M0' - PRINT FINAL MEMBER STRESSES (DEFAULT) 'M1' - PRINT MEMBER STRESSES AT EACH LOAD INCREMENT 'M2' - PRINT MEMBER STRESSES AT EACH LOOP 'MP' - OPTION TO ONLY INCLUDE PLASTIC MEMBERS IN MEMBER INTERNAL LOADS REPORT 'SP' - PRINT MEMBER STRESSES AT EACH SUBAREA AROUND THE CROSS SECTION JOINT STRENGTH REPORT OPTION: 'J1' - PRINT JOINT STRENGTH AT EACH LOAD INCREMENT 'J2' - PRINT JOINT STRENGTH AT EACH LOOP SUMMARY REPORT OPTION: 'SM' - COLLAPSE SUMMARY REPORT 'MS' - MEMBER SUMMARY REPORT OPTION 'PW' - OPTION TO PRINT MEMBER WARNING MESSAGES
(26-27)	PILEHEAD REACTION REPORT OPTION: 'F0' - PRINT FINAL REACTIONS 'F1' - PRINT REACTIONS AT EACH LOAD INCREMENT 'F2' - PRINT REACTIONS AT EACH LOOP
(30-31)	ENTER 'VM' FOR A VON MISES STRESS CHECK FOR PLATES DESIGNATED AS ELASTIC.

COLUMNS	COMMENTARY
(32-36)	ENTER THE PLASTICITY RATIO FOR THE MEMBER STRESS REPORT. ONLY THOSE MEMBER SEGMENTS THAT EXCEED THIS LEVEL WILL BE INCLUDED IN THIS REPORT.
(38-42)	ENTER THE PLASTICITY RATIO FOR THE PILE DETAIL REPORT. ONLY THOSE PILE INCREMENTS THAT EXCEED THIS LEVEL WILL BE INCLUDED IN THIS REPORT.
(44-48)	ENTER THE PLASTICITY RATIO FOR THE PLATE STRESS DETAIL REPORT. ONLY THOSE MEMBER SEGMENTS THAT EXCEED THIS LEVEL WILL BE INCLUDED IN THIS REPORT.
(50-51)	ENTER 'ES' TO PRODUCE A PLATE STRAIN REPORT. THE REPORT IS GENERATED WHENEVER A PLATE STRESS REPORT IS GENERATED.
(52-53)	ENTER 'ER' TO PRODUCE A PLATE STRAIN RATE REPORT. THE REPORT IS GENERATED AT THE END OF EACH LOAD INCREMENT.
(54-55)	ENTER 'PS' TO PRODUCE A PLATE PRINCIPAL STRAIN REPORT. THE REPORT IS GENERATED WHENEVER A PLATE STRESS REPORT IS
(56-57)	ENTER 'PR' TO PRODUCE A PLATE PRINCIPAL STRAIN RATE REPORT. THE REPORT IS GENERATED AT THE END OF EACH LOAD INCREMENT.

LINE LABEL	OUTPUT SELECTIONS											VON MISES PLATE CHECK	MEMBER DETAIL REPORT PLASTIC RATIO	PILE DETAIL REPORT PLASTIC RATIO	PLATE DETAIL REPORT PLASTIC RATIO	PLATE STRAIN REPORT SELECTIONS				LEAVE BLANK
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH					STRAIN	STRAIN RATE	PRINCIPAL STRAIN	PRINCIPAL STRAIN RATE	
CLPRPT																				
1-- 6	8-- 9	10--11	12--13	14--15	16--17	18--19	20--21	22--23	24--25	26--27	28--29	30--31	32--36	38--42	44--48	50--51	52--53	54--55	56--57	58--80

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE SIGNIFIES THE END OF THE COLLAPSE INPUT AND IS THE LAST LINE OF THE COLLAPSE INPUT.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

SHIP IMPACT ENERGY

COLUMNS	COMMENTARY
GENERAL	THIS LINE SPECIFIES THE TOTAL IMPACT ENERGY IN A SHIP IMPACT. IT PROVIDES A MEANS FOR THE USER TO SPECIFY SHIP VELOCITY AND MASS WITH THE PROGRAM COMPUTING IMPACT ENERGY.
(8-15)	ENTER THE SHIP MASS M.
(17-22)	ENTER THE ADDED MASS COEFFICIENT C. SHIP KINETIC ENERGY IS CALCULATED WITH THE FORMULA $E_k = 1/2 * C * M * V^2$.
(24-29)	ENTER THE SHIP VELOCITY V.

LINE LABEL	SHIP MASS	ADDED MASS COEFFICIENT	SHIP VELOCITY	LEAVE BLANK
ENERGY				
1-- 6	8<--15	17<--22	24<--29	30----80
DEFAULT				
ENGLISH	TON		FT/S	
METRIC	TONNE		M/S	

GROUTED MEMBER YIELD STRESS MODIFICATION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY THE YIELD STRESS FOR GROUTED TUBULAR MEMBERS. THE LINE MAY BE APPLIED ONLY TO CROSS SECTIONS WITH THE 'TUB' TYPE WHICH HAVE A SPECIFIED INNER TUBE. IF APPLIED TO MEMBERS WITHOUT THE 'TUB' CROSS SECTION TYPE, OR TO MEMBERS WITH THE 'TUB' CROSS SECTION TYPE BUT WITHOUT AN INNER TUBE, THIS LINE IS IGNORED.
(8-12)	ENTER THE YIELD STRESS. THIS YIELD STRESS IS APPLIED ONLY TO THE INNER (GROUTED) TUBE OF THE 'TUB' CROSS SECTION.
(14-72)	ENTER THE CONNECTING JOINT NAMES OF UP TO SIX GROUTED TUBULAR MEMBERS SELECTED FOR GROUTED YIELD STRESS MODIFICATION. USE AS MANY 'GRMSEL' INPUT LINES AS REQUIRED.

[illegible]

ELASTIC MEMBER GROUPS

COLUMNS	COMMENTARY
GENERAL	MEMBERS OF THESE GROUPS ARE TO BE CONSIDERED AS ELASTIC LARGE DEFLECTION ELEMENTS WITH NO PLASTIC OR BUCKLING EFFECTS INCLUDED. THIS CAN SIGNIFICANTLY REDUCE THE TIME REQUIRED TO PERFORM AN ANALYSIS.
(1- 6)	ENTER 'GRPDEL'.
(16-18)	ENTER THE FIRST GROUP IDENTIFIER. THIS IDENTIFIER MUST CORRESPOND TO A GROUP IDENTIFIER ON A SACS IV 'GRUP' LINE.
(20-74)	THE REMAINING GROUP IDENTIFIER FIELDS ARE SIMILAR. ANY NUMBER OF 'GRPDEL' INPUT LINES CAN BE SPECIFIED.

LINE LABEL	MEMBER GROUP IDENTIFIERS														
	GROUP ID 1	GROUP ID 2	GROUP ID 3	GROUP ID 4	GROUP ID 5	GROUP ID 6	GROUP ID 7	GROUP ID 8	GROUP ID 9	GROUP ID 10	GROUP ID 11	GROUP ID 12	GROUP ID 13	GROUP ID 14	GROUP ID 15
GRPDEL															
1-- 6	16--18	20--22	24--26	28--30	32--34	36--38	40--42	44--46	48--50	52--54	56--58	60--62	64--66	68--70	72--74

ELASTIC MEMBER GROUPS

COLUMNS	COMMENTARY
GENERAL	MEMBERS OF THESE GROUPS ARE TO BE CONSIDERED AS ELASTIC LARGE DEFLECTION ELEMENTS WITH NO PLASTIC OR BUCKLING EFFECTS INCLUDED. THIS CAN SIGNIFICANTLY REDUCE THE TIME REQUIRED TO PERFORM AN ANALYSIS.
(1- 6)	ENTER 'GRPELA'.
(16-18)	ENTER THE FIRST GROUP IDENTIFIER. THIS IDENTIFIER MUST CORRESPOND TO A GROUP IDENTIFIER ON A SACS IV 'GRUP' LINE.
(20-74)	THE REMAINING GROUP IDENTIFIER FIELDS ARE SIMILAR. ANY NUMBER OF 'GRPELA' INPUT LINES CAN BE SPECIFIED.

LINE LABEL	MEMBER GROUP IDENTIFIERS														
	GROUP ID 1	GROUP ID 2	GROUP ID 3	GROUP ID 4	GROUP ID 5	GROUP ID 6	GROUP ID 7	GROUP ID 8	GROUP ID 9	GROUP ID 10	GROUP ID 11	GROUP ID 12	GROUP ID 13	GROUP ID 14	GROUP ID 15
GRPELA															
1-- 6	16--18	20--22	24--26	28--30	32--34	36--38	40--42	44--46	48--50	52--54	56--58	60--62	64--66	68--70	72--74

MEMBER GROUP SUBSEGMENT SPECIFICATION

COLUMNS	COMMENTARY
GENERAL	THE FOLLOWING MEMBER GROUPS CONTAIN MEMBERS THAT ARE TO BE DIVIDED INTO A SPECIFIED NUMBER OF SUBSEGMENTS FOR THE PURPOSE OF AN ELASTO-PLASTIC ANALYSIS. THIS NUMBER OVERRIDES THE DEFAULT SETTING FOR THE NUMBER OF SUBSEGMENTS FROM COLUMNS 14-16 OF THE CLPOPT LINE.
(1- 6)	ENTER 'GRPSEG' ON ALL INPUT LINES IN THIS SET.
(8- 9)	THE NUMBER OF SUBSEGMENTS TO BE USED FOR EACH MEMBER THAT IS CONTAINED BY A MEMBER GROUP ON THE INPUT LINE.
(14-16)	ENTER MEMBER GROUP NAME.
(18-72)	REPEAT FOR ADDITIONAL MEMBER GROUPS. FIFTEEN MEMBER GROUPS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBER GROUPS.

LINE LABEL	NO. OF SUBSEGS	MEMBER GROUP IDENTIFIERS														
		GROUP ID 1	GROUP ID 2	GROUP ID 3	GROUP ID 4	GROUP ID 5	GROUP ID 6	GROUP ID 7	GROUP ID 8	GROUP ID 9	GROUP ID 10	GROUP ID 11	GROUP ID 12	GROUP ID 13	GROUP ID 14	GROUP ID 15
GRPSEG																
1-- 6	8--> 9	14--16	18--20	22--24	26--28	30--32	34--36	38--40	42--44	46--48	50--52	54--56	58--60	62--64	66--68	70--72

SKIPPED LOCAL BUCKLING FOR MEMBER GROUPS

COLUMNS	COMMENTARY
GENERAL	MEMBERS OF THESE GROUPS ARE TO BE CONSIDERED AS ELASTO-PLASTIC LARGE DEFLECTION ELEMENTS WITH NO LOCAL BUCKLING EFFECTS INCLUDED. ANY LOCAL BUCKLING CHECK SPECIFIED UNDER THE 'LB' OPTION WILL BE SKIPPED FOR MEMBERS FROM THESE GROUPS.
(1- 6)	ENTER 'GRPSKP'.
(16-18)	ENTER THE FIRST GROUP IDENTIFIER. THIS IDENTIFIER MUST CORRESPOND TO A GROUP IDENTIFIER ON A SACS IV 'GRUP' LINE.
(20-74)	THE REMAINING GROUP IDENTIFIER FIELDS ARE SIMILAR. ANY NUMBER OF 'GRPSKP' INPUT LINES CAN BE SPECIFIED.

LINE LABEL	MEMBER GROUP IDENTIFIERS														
	GROUP ID 1	GROUP ID 2	GROUP ID 3	GROUP ID 4	GROUP ID 5	GROUP ID 6	GROUP ID 7	GROUP ID 8	GROUP ID 9	GROUP ID 10	GROUP ID 11	GROUP ID 12	GROUP ID 13	GROUP ID 14	GROUP ID 15
GRPSKP															
1-- 6	16--18	20--22	24--26	28--30	32--34	36--38	40--42	44--46	48--50	52--54	56--58	60--62	64--66	68--70	72--74

JOINT FLEXIBILITY JOINT SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE ALLOWS THE USER TO INCLUDE OR EXCLUDE JOINTS WHEN CONSIDERING JOINT FLEXIBILITY. ALL BRACES CONNECTED TO THE SPECIFIED JOINTS ARE INCLUDED OR EXCLUDED PROVIDED ONE OF THE JOINT FLEXIBILITY OPTIONS IS SPECIFIED ON THE CLPOPT LINE. THIS LINE IS IGNORED IF NO FLEXIBILITY OPTION IS SPECIFIED ON THE CLPOPT LINE.
(1- 5)	ENTER 'JFSEL'.
(7)	ENTER 'I' TO INCLUDE OR 'X' TO EXCLUDE THE JOINTS SPECIFIED.
NOTE:	THE INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND CANNOT BE USED TOGETHER. ALL JOINTS SPECIFIED ON JFSEL LINES MUST BE EITHER INCLUDED OR EXCLUDED.
(9-12)	ENTER THE NAME OF THE FIRST JOINT TO BE INCLUDED OR EXCLUDED.
(14-77)	ENTER ADDITIONAL JOINTS TO BE INCLUDED OR EXCLUDED.

LINE LABEL	INCLUDE OR EXCLUDE	JOINT NAMES													
		1ST JOINT	2ND JOINT	3RD JOINT	4TH JOINT	5TH JOINT	6TH JOINT	7TH JOINT	8TH JOINT	9TH JOINT	10TH JOINT	11TH JOINT	12TH JOINT	13TH JOINT	14TH JOINT
JFSEL															
1-- 5	7	9-->12	14-->17	19-->22	24-->27	29-->32	34-->37	39-->42	44-->47	49-->52	54-->57	59-->62	64-->67	69-->72	74-->77

JOINT STRENGTH OPTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY JOINT STRENGTH OPTIONS. IF OMITTED, THEN DEFAULT OPTIONS WILL BE USED.
(11)	IF THE BRACE LOADS ARE TO BE BACKED OFF TO THE CHORD OUTER SURFACE, ENTER 'R' HERE. OTHERWISE, THE PROGRAM WILL USE THE BRACE END FORCES.
(12-17)	ENTER THE MINIMUM GAP ALLOWED FOR THE JOINT STRENGTH ANALYSIS.
(18-23)	ENTER THE MAXIMUM GAP ALLOWED FOR THE JOINT STRENGTH ANALYSIS.
(25)	USE EFFECTIVE THICKNESS FOR GROUTED ELEMENTS. ENTER '1' FOR EFF THICK BASED ON THE COMPOSITE SECTION MOMENT OF INERTIA OR '2' FOR EFF THICK BASED ON MOMENT OF INERTIAS OF THE TWO WALLS OR '3' FOR EFF THICK BASED ON SQUARE ROOT OF SUM OF SQUARES (SRSS) OF WALL THICKNESSES.
(26-30)	ENTER THE EFFECTIVE THICKNESS LIMIT EXPRESSED AS A FACTOR OF THE WALL THICKNESS OF THE LARGER (OUTSIDE) TUBE.
(31-35)	ENTER THE UNITY CHECK LOWER LIMIT. ONLY JOINTS WITH STRENGTH UNITY CHECK RATIOS ABOVE THIS VALUE WILL BE REPORTED.
(37)	PERFORM THE ISO 19902 JOINT STRENGTH CHECK USING BRACE UTILIZATION. ENTER 'U' TO ASSUME A BRACE UTILIZATION OF UNITY, ENTER A 'B' TO CALCULATE THE BRACE-END UTILIZATION USING ISO 19902 EQUATIONS 13.3-2 AND 13.3-8.

LINE LABEL	RELIEF OPTION	MINIMUM GAP	MAXIMUM GAP	EFFECTIVE THICKNESS OPTION	EFFECTIVE THICKNESS LIMIT RATIO	PRINT UC LEVEL	BRACE UTILIZATION OPTION	LEAVE BLANK
JSOPT								
1-- 5	11	12<--17	18<--23	25	26<--30	31--35	37	38--80
DEFAULT		-100.0 ENGL	+1000.0 ENGL		1.75			
ENGLISH		IN	IN					
METRIC		CM	CM					

JOINT STRENGTH JOINT SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE ALLOWS THE USER TO INCLUDE OR EXCLUDE JOINTS WHEN CONSIDERING JOINT STRENGTH. ALL BRACES CONNECTED TO THE SPECIFIED JOINTS ARE INCLUDED OR EXCLUDED PROVIDED ONE OF THE JOINT STRENGTH OPTIONS IS SPECIFIED ON THE CLPOPT LINE. THIS LINE IS IGNORED IF NO STRENGTH OPTION IS SPECIFIED ON THE CLPOPT LINE.
(1- 5)	ENTER 'JSSEL'.
(7)	ENTER 'I' TO INCLUDE OR 'X' TO EXCLUDE THE JOINTS SPECIFIED.
NOTE:	THE INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND CANNOT BE USED TOGETHER. ALL JOINTS SPECIFIED ON JSSEL LINES MUST BE EITHER INCLUDED OR EXCLUDED.
(9-12)	ENTER THE NAME OF THE FIRST JOINT TO BE INCLUDED OR EXCLUDED.
(14-77)	ENTER ADDITIONAL JOINTS TO BE INCLUDED OR EXCLUDED.

LINE LABEL	INCLUDE OR EXCLUDE	JOINT NAMES													
		1ST JOINT	2ND JOINT	3RD JOINT	4TH JOINT	5TH JOINT	6TH JOINT	7TH JOINT	8TH JOINT	9TH JOINT	10TH JOINT	11TH JOINT	12TH JOINT	13TH JOINT	14TH JOINT
JSSEL															
1-- 5	7	9-->12	14-->17	19-->22	24-->27	29-->32	34-->37	39-->42	44-->47	49-->52	54-->57	59-->62	64-->67	69-->72	74-->77

JOINT SELECTION INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SELECT JOINTS FOR DEFLECTION PRINT. IF OMITTED, THEN ALL JOINTS ARE SELECTED BY DEFAULT.
(12-80)	ENTER THE JOINT NAMES OF JOINTS SELECTED FOR OUTPUT DEFLECTION PRINT. USE AS MANY OF THESE INPUT LINES AS DESIRED.

LINE LABEL	JOINT SELECTIONS FOR DEFLECTION PRINT													
	1ST JOINT	2ND JOINT	3RD JOINT	4TH JOINT	5TH JOINT	6TH JOINT	7TH JOINT	8TH JOINT	9TH JOINT	10TH JOINT	11TH JOINT	12TH JOINT	13TH JOINT	14TH JOINT
JTSEL														
1-- 5	12-->15	17-->20	22-->25	27-->30	32-->35	37-->40	42-->45	47-->50	52-->55	57-->60	62-->65	67-->70	72-->75	77-->80

LOADING SEQUENCE INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE CAN BE USED IN ANY COLLAPSE ANALYSIS RUN. HOWEVER, IT IS REQUIRED FOR ANY DYNAMIC COLLAPSE ANALYSIS. IT IS USED TO SPECIFY THE LOAD STEPS IN A LOAD SEQUENCE AND ALSO THE TIME DURATION FOR THE LOAD STEPS. IN A NONLINEAR ANALYSIS, THE ORDER IN WHICH LOADS ARE APPLIED CAN BE SIGNIFICANT. FOR EXAMPLE, THE DEAD LOAD SHOULD BE APPLIED BEFORE ANY ENVIRONMENTAL LOADING. AS MANY AS SIX LOAD SEQUENCES CAN BE DEFINED. EACH OF THESE WILL BE ANALYZED AS INDEPENDENT NONLINEAR ANALYSES. A TOTAL OF 50 LOAD PATHS ARE ALLOWED.
(7-10)	ENTER THE IDENTIFICATION OF THIS LOAD SEQUENCE. EACH LOAD SEQUENCE MUST HAVE A NON-BLANK LOAD SEQUENCE IDENTIFIER. IF THIS FIELD IS LEFT BLANK, THE LOAD PATHS ARE APPLIED TO THE PREVIOUS LOAD SEQUENCE.
(21-24)	ENTER THE SACS IV LOAD CASE NAME FOR THE FIRST LOAD TO BE APPLIED.
(25-29)	ENTER THE NUMBER OF INCREMENTS FOR THIS LOAD STEP. THIS IS THE NUMBER OF STEPS FROM THE STARTING LOAD FACTOR TO THE ENDING LOAD FACTOR. IF THE STARTING FACTOR IS GREATER THAN ZERO, THEN AN ADDITIONAL LOAD STEP IS CREATED TO REACH THE STARTING LOAD FACTOR POSITION.
(30-36)	ENTER THE STARTING LOAD FACTOR. THIS FACTOR CAN BE USED TO SKIP THE LINEAR PORTION ON THE ANALYSIS AND SAVE UNNECESSARY RUN TIME.
(37-43)	ENTER THE ENDING LOAD FACTOR. THIS FACTOR MUST BE GREATER THAN OR EQUAL TO THE STARTING FACTOR.
(44-50)	ENTER THE TIME DURATION FOR THIS LOAD STEP. THIS IS REQUIRED FOR A DYNAMIC ANALYSIS.

LINE LABEL	LOAD SEQUENCE ID	LOAD CASE NAME	NUMBER OF INCREMENTS	STARTING FACTOR	ENDING FACTOR	TIME DURATION	LEAVE BLANK
LDAPL							
1-- 5	7--10	21-->24	25-->29	30<--36	37<--43	44<--50	51-----80

LOADING SEQUENCE INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY THE LOAD SEQUENCE. IN A NONLINEAR ANALYSIS, THE ORDER IN WHICH LOADS ARE APPLIED CAN BE SIGNIFICANT. FOR EXAMPLE, THE DEAD LOAD SHOULD BE APPLIED BEFORE ANY ENVIRONMENTAL LOADING. AS MANY AS SIX LOAD SEQUENCES CAN BE DEFINED. EACH OF THESE WILL BE ANALYZED AS INDEPENDENT NONLINEAR ANALYSES. A TOTAL OF 50 LOAD PATHS ARE ALLOWED.
(7-10)	ENTER THE IDENTIFICATION OF THIS LOAD SEQUENCE. EACH LOAD SEQUENCE MUST HAVE A NON-BLANK LOAD SEQUENCE IDENTIFIER. IF THIS FIELD IS LEFT BLANK, THE LOAD PATHS ARE APPLIED TO THE PREVIOUS LOAD SEQUENCE.
(21-24)	ENTER THE SACS IV LOAD CASE NAME FOR THE FIRST LOAD TO BE APPLIED.
(25-28)	ENTER THE NUMBER OF INCREMENTS FOR THIS LOAD STEP. THIS IS THE NUMBER OF STEPS FROM THE STARTING LOAD FACTOR TO THE ENDING LOAD FACTOR. IF THE STARTING FACTOR IS GREATER THAN ZERO, THEN AN ADDITIONAL LOAD STEP IS CREATED TO REACH THE STARTING LOAD FACTOR POSITION.
(29-34)	ENTER THE STARTING LOAD FACTOR. THIS FACTOR CAN BE USED TO SKIP THE LINEAR PORTION ON THE ANALYSIS AND SAVE UNNECESSARY RUN TIME.
(35-40)	ENTER THE ENDING LOAD FACTOR. THIS FACTOR MUST BE GREATER THAN OR EQUAL TO THE STARTING FACTOR.
(41-60)	ENTER THE PARAMETERS FOR THE SECOND LOAD STEP.
(61-80)	ENTER THE PARAMETERS FOR THE THIRD LOAD STEP.
	NOTE: IF MORE THAN 3 LOAD STEPS ARE DESIRED, THEY MAY BE ENTERED ON ADDITIONAL LDSEQ INPUT LINES LEAVING THE LOAD SEQUENCE IDENTIFIER BLANK.

LINE LABEL	LOAD SEQUENCE ID	FIRST LOAD STEP				SECOND LOAD STEP				THIRD LOAD STEP			
		LOAD CASE NAME	NUMBER OF INCREMENTS	STARTING FACTOR	ENDING FACTOR	LOAD CASE NAME	NUMBER OF INCREMENTS	STARTING FACTOR	ENDING FACTOR	LOAD CASE NAME	NUMBER OF INCREMENTS	STARTING FACTOR	ENDING FACTOR
LDSEQ													
1-- 5	7--10	21-->24	25-->28	29<--34	35<--40	41-->44	45-->48	49<--54	55<--60	61-->64	65-->68	69<--74	75<--80

MATERIAL MODEL ASSIGNMENT TO MEMBER GROUPS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO ASSIGN A MATERIAL MODEL (POST-YILED STRESS-STRAIN BEHAVIOR) TO DIFFERENT MEMBER GROUPS AVAILABLE IN SACS INPUT FILE. IF A MEMBER IS NOT ASSIGNED ANY MATERIAL MODEL, IT WILL HAVE A CONSTANT STRAIN-HARDEDNING RATIO AS DEFINED IN THE 'CLPOPT' LINE.
(1- 6)	ENTER 'MATGRP'.
(8-11)	ENTER THE MATERIAL MODEL NAME. MAXIMUM 50 MATERIAL MODELS CAN BE DEFINED IN A COLLAPSE INPUT FILE.
(13-15)	ENTER 'ALL' IF ALL THE MEMBERS ARE TO BE ASSIGNED THIS MATERIAL MODEL. ONLY THE FIRST 'ALL' ASSIGNMENT WILL BE CONSIDERED. ALL OTHER MATERIAL MODEL ASSIGNMENTS WILL BE IGNORED.
(17-75)	ENTER MEMBER GROUP IDENTIFIERS. MAXIMUM OF 15 MEMBER GROUPS CAN BE SPECIFIED IN A LINE. REPEAT THE 'MATGRP' CARD FOR ASSIGNING MORE MEMBER GROUPS TO THE SAME MATERIAL MODEL. 'MATGRP' CARDS SHOULD BE FOLLOWED BY 'MATPRP HEAD' AND 'MATPRP PLAS' CARDS, WHICH ARE USED TO DEFINE THE MATERIAL MODEL.

LINE LABEL	MATERIAL MODEL NAME	SELECT 'ALL'	MEMBER GROUP IDENTIFIERS														
			GROUP ID 1	GROUP ID 2	GROUP ID 3	GROUP ID 4	GROUP ID 5	GROUP ID 6	GROUP ID 7	GROUP ID 8	GROUP ID 9	GROUP ID 10	GROUP ID 11	GROUP ID 12	GROUP ID 13	GROUP ID 14	GROUP ID 15
MATGRP																	
1-- 6	8-->11	13--15	17--19	21--23	25--27	29--31	33--35	37--39	41--43	45--47	49--51	53--55	57--59	61--63	65--67	69--71	73--75

MATERIAL PROPERTY HEADER LINE

COLUMNS	COMMENTARY
GENERAL	USE THIS LINE TO START MATERIAL MODEL DEFINITION. THIS LINE MUST BE FOLLOWED BY 'MATPRP PLAS' LINE(S).
(1- 6)	ENTER 'MATPRP'.
(8-11)	ENTER 'HEAD'.
(13-16)	ENTER THE NAME OF MATERIAL MODEL. IT SHOUDL BE SAME AS THE ONE SPECIFIED IN THE PRECEDING 'MATGRP' LINE.
(17-80)	SHOULD BE LEFT BLANK.
	THIS LINE MUST BE FOLLOWED BY 'MATPRP PLAS' LINE(S) TO DEFINE THE POST YIELD BEHAVIOR FOR THIS MATERIAL MODEL.

LINE LABEL	HEAD LABEL	MATERIAL MODEL NAME	
MATPRP	HEAD		
1-- 6	8--11	13--16	17-----80

PLASTIC STRESS-STRAIN INPUT LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INPUT THE POST YIELD PLASTIC STRAIN - STRESS FACTOR DATA FOR MATERIAL MODELS SPECIFIED USING 'MATGRP' LINES. THE FIRST DATA POINT IN THE FIRST 'MATPRP PLAS' CARD REPRESENTS THE YIELD POINT AND MUST BE ENTERED AS (0.0, 1.0). THE PLASTIC STRAIN AND THE STRESS FACTOR VALUES MUST INCREASE MONOTONICALLY THEREAFTER. FOR VALUES OF PLASTIC STRAIN GREATER THAN THE LARGEST SPECIFIED VALUE, THE VALUE OF STRESS FACTOR IS CALCULATED USING THE DEFAULT STRAIN HARDENING RATIO SPECIFIED IN THE CLPOPT LINE.
(1- 6)	ENTER 'MATPRP'.
(8-11)	ENTER 'PLAS'.
(13-72)	ENTER PLASTIC STRAIN AND STRESS FACTOR VALUES FOR EACH DATA POINT. PLASTIC STRAIN IS EQUAL TO (MECHANICAL STRAIN - YIELD STRAIN). STRESS FACTOR IS EQUAL TO (TOTAL STRESS / YIELD STRESS).
	THIS LINE MAY BE REPEATED AS NECESSARY. MAXIMUM 50 DATA POINTS ARE ALLOWED FOR EACH MATERIAL MODEL.

[illegible]

ELASTIC MEMBER INPUT

COLUMNS	COMMENTARY
GENERAL	THE FOLLOWING MEMBERS ARE TO BE CONSIDERED AS ELASTIC LARGE DEFLECTION ELEMENTS WITH NO PLASTIC OR BUCKLING EFFECTS INCLUDED. THIS CAN SIGNIFICANTLY REDUCE THE RUN TIME REQUIRED TO PERFORM AN ANALYSIS.
(1- 6)	ENTER 'MEMDEL' ON ALL INPUT LINES IN THIS SET.
(9-12)	ENTER JOINT 1 FOR THE FIRST MEMBER.
(14-17)	ENTER JOINT 2 FOR THE FIRST MEMBER.
(20-72)	REPEAT FOR ADDITIONAL MEMBERS. SIX MEMBERS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBERS.

LINE LABEL	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER	
	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2
MEMDEL												
1-- 6	9-->12	14-->17	20-->23	25-->28	31-->34	36-->39	42-->45	47-->50	53-->56	58-->61	64-->67	69-->72

DUCTILITY LIMIT FOR AN INDIVIDUAL MEMBER

COLUMNS	COMMENTARY
GENERAL	EACH OF THE FOLLOWING MEMBERS ARE TO BE REMOVED FROM THE ANALYSIS WHEN THE SPECIFIED TENSILE DUCTILITY LIMIT HAS BEEN REACHED. FOR ALL SUBSEQUENT INCREMENTS, EACH SPECIFIED MEMBER IS NO LONGER CONSIDERED TO BE PART OF THE STUCTURE. THE DUCILITY LIMIT IS EXPRESSED AS A PRECENTAGE. THE INDIVIDUAL MEMBER DUCTILITY LIMIT OVERRIDES THE GLOBAL MEMBER DUCTILITY LIMIT THAT IS SPECIFIED ON THE CLPOP2 LINE.
(1- 6)	ENTER 'MEMDUC' ON ALL INPUT LINES IN THIS SET.
(8-12)	PERCENTAGE OF DUCTILITY AT WHICH THE MEMBER IS TO BE REMOVED.
(14-17)	ENTER JOINT 1 FOR THE FIRST MEMBER.
(19-22)	ENTER JOINT 2 FOR THE FIRST MEMBER.
(24-72)	REPEAT FOR ADDITIONAL MEMBERS. SIX MEMBERS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBERS.

LINE LABEL	DUCTILITY	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER	
	LIMIT (%)	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2
MEMDUC													
1-- 6	8-->12	14-->17	19-->22	24-->27	29-->32	34-->37	39-->42	44-->47	49-->52	54-->57	59-->62	64-->67	69-->72

ELASTIC MEMBER INPUT

COLUMNS	COMMENTARY
GENERAL	THE FOLLOWING MEMBERS ARE TO BE CONSIDERED AS ELASTIC LARGE DEFLECTION ELEMENTS WITH NO PLASTIC OR BUCKLING EFFECTS INCLUDED. THIS CAN SIGNIFICANTLY REDUCE THE RUN TIME REQUIRED TO PERFORM AN ANALYSIS.
(1- 6)	ENTER 'MEMELA' ON ALL INPUT LINES IN THIS SET.
(9-12)	ENTER JOINT 1 FOR THE FIRST MEMBER.
(14-17)	ENTER JOINT 2 FOR THE FIRST MEMBER.
(20-72)	REPEAT FOR ADDITIONAL MEMBERS. SIX MEMBERS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBERS.

LINE LABEL	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER	
	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2
MEMELA												
1-- 6	9-->12	14-->17	20-->23	25-->28	31-->34	36-->39	42-->45	47-->50	53-->56	58-->61	64-->67	69-->72

MEMBER REMOVAL

COLUMNS	COMMENTARY
GENERAL	THE FOLLOWING MEMBERS ARE TO BE REMOVED FROM THE ANALYSIS ON COMPLETION OF A SPECIFIED LOAD INCREMENT. FOR ALL SUBSEQUENT INCREMENTS, EACH SPECIFIED MEMBER IS NO LONGER CONSIDERED TO BE PART OF THE STUCTURE.
(1- 6)	ENTER 'MEMREM' ON ALL INPUT LINES IN THIS SET.
(8-12)	LOAD INCREMENT AT WHICH THE MEMBER IS TO BE REMOVED.
(14-17)	ENTER JOINT 1 FOR THE FIRST MEMBER.
(19-22)	ENTER JOINT 2 FOR THE FIRST MEMBER.
(24-72)	REPEAT FOR ADDITIONAL MEMBERS. SIX MEMBERS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBERS.

LINE LABEL	LOAD	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER	
	INCREMENT	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2
MEMREM													
1-- 6	8-->12	14-->17	19-->22	24-->27	29-->32	34-->37	39-->42	44-->47	49-->52	54-->57	59-->62	64-->67	69-->72

MEMBER SUBSEGMENT SPECIFICATION

COLUMNS	COMMENTARY
GENERAL	THE FOLLOWING MEMBERS ARE TO BE DIVIDED INTO A SPECIFIED NUMBER OF SUBSEGMENTS FOR THE PURPOSE OF AN ELASTO-PLASTIC ANALYSIS. THIS NUMBER OVERRIDES THE DEFAULT SETTING FOR THE NUMBER OF SUBSEGMENTS FROM COLUMNS 14-16 OF THE CLPOPT LINE, AS WELL AS SUBSEGMENT OVERRIDES THAT ARE SPECIFIED USING THE GRPSEG LINE.
(1- 6)	ENTER 'MEMSEG' ON ALL INPUT LINES IN THIS SET.
(8- 9)	THE NUMBER OF SUBSEGMENTS TO BE USED FOR EACH MEMBER ON THE INPUT LINE.
(14-17)	ENTER JOINT 1 FOR THE FIRST MEMBER.
(19-22)	ENTER JOINT 2 FOR THE FIRST MEMBER.
(24-72)	REPEAT FOR ADDITIONAL MEMBERS. SIX MEMBERS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBERS.

LINE LABEL	NUMBER OF SUBSEGMENTS	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER	
		JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2
MEMSEG													
1-- 6	8--> 9	14-->17	19-->22	24-->27	29-->32	34-->37	39-->42	44-->47	49-->52	54-->57	59-->62	64-->67	69-->72

MEMBER SELECTION INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SELECT MEMBERS FOR OUTPUT PRINT. IF OMITTED, THEN ALL MEMBERS ARE SELECTED BY DEFAULT.
(12-80)	ENTER THE CONNECTING JOINT NAMES OF MEMBERS SELECTED FOR OUTPUT PRINT. USE AS MANY OF THESE INPUT LINES AS DESIRED.

LINE LABEL	MEMBER SELECTIONS FOR OUTPUT REPORTS													
	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER		7TH MEMBER	
	1ST JOINT	2ND JOINT	1ST JOINT	2ND JOINT	1ST JOINT	2ND JOINT	1ST JOINT	2ND JOINT	1ST JOINT	2ND JOINT	1ST JOINT	2ND JOINT	1ST JOINT	2ND JOINT
MEMSEL														
1-- 6	12-->15	17-->20	22-->25	27-->30	32-->35	37-->40	42-->45	47-->50	52-->55	57-->60	62-->65	67-->70	72-->75	77-->80

SKIPPED MEMBER LOCAL BUCKLING INPUT

COLUMNS	COMMENTARY
GENERAL	THE FOLLOWING MEMBERS ARE TO BE CONSIDERED AS ELASTO-PLASTIC LARGE DEFLECTION ELEMENTS WITH NO LOCAL BUCKLING EFFECTS INCLUDED. ANY LOCAL BUCKLING CHECK SPECIFIED UNDER THE 'LB' OPTION WILL BE SKIPPED FOR THESE MEMBERS.
(1- 6)	ENTER 'MEMSKP' ON ALL INPUT LINES IN THIS SET.
(9-12)	ENTER JOINT 1 FOR THE FIRST MEMBER.
(14-17)	ENTER JOINT 2 FOR THE FIRST MEMBER.
(20-72)	REPEAT FOR ADDITIONAL MEMBERS. SIX MEMBERS CAN BE INPUT PER LINE. REPEAT AS REQUIRED FOR ADDITIONAL MEMBERS.

LINE LABEL	1ST MEMBER		2ND MEMBER		3RD MEMBER		4TH MEMBER		5TH MEMBER		6TH MEMBER	
	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2	JOINT 1	JOINT 2
MEMSKP												
1-- 6	9-->12	14-->17	20-->23	25-->28	31-->34	36-->39	42-->45	47-->50	53-->56	58-->61	64-->67	69-->72

MSL JOINT FLEXIBILITY OPTIONS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY PARAMETERS WHEN USING JOINT FLEXIBILITY OPTIONS FROM MSL.
(8- 9)	SELECT THE FLEXIBILITY OPTION: 'MF' - MEAN LEVEL 'CF' - CHARACTERISTIC LEVEL
(10-11)	SELECT THE STRENGTH CHECK OPTION: 'MS' - MEAN LEVEL 'CS' - CHARACTERISTIC LEVEL
(12-13)	SELECT THE FRACTURE CHECK OPTION: 'MT' - MEAN LEVEL 'CT' - CHARACTERISTIC LEVEL
(15-19)	ENTER THE JOINT DISTORTION TOLERANCE.
(20-24)	ENTER THE JOINT ROTATION TOLERANCE.

LINE LABEL	JOINT FLEXIBILITY	JOINT STRENGTH	JOINT FRACTURE	TOLERANCES		LEAVE BLANK
				DISTORTION TOLERANCE	ROTATION TOLERANCE	
MSLOPT						
1-- 6	8-- 9	10--11	12--13	15<--19	20<--24	25<-----80
DEFAULT				0.001 ENGL	0.001	
ENGLISH				IN	RAD	
METRIC				CM	RAD	

JOINT TO JOINT NONLINEAR SPRING INPUT

COLUMNS	COMMENTARY																
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY NONLINEAR SPRINGS BETWEEN JOINTS. ANY NUMBER OF NONLINEAR SPRINGS CAN BE USED.																
(8-11)	ENTER THE BEGIN JOINT OF THE SPRING.																
(12-15)	ENTER THE END JOINT OF THE SPRING.																
(18-19)	ENTER THE DEGREE OF FREEDOM FOR THIS CONSTRAINT FROM THE FOLLOWING: 'DX' - DEFLECTION IN LOCAL X-DIRECTION 'DY' - DEFLECTION IN LOCAL Y-DIRECTION 'DZ' - DEFLECTION IN LOCAL Z-DIRECTION 'RX' - ROTATION ABOUT LOCAL X-AXIS 'RY' - ROTATION ABOUT LOCAL Y-AXIS 'RZ' - ROTATION ABOUT LOCAL Z-AXIS NOTE THAT LOCAL COORDINATES ARE DEFINED IN THE SAME MANNER AS FOR MEMBERS.																
(25-80)	ENTER THE FORCE-DEFLECTION OR MOMENT-ROTATION PAIRS OF VALUES IN ORDER OF INCREASING DEFLECTION/ROTATION USING THE UNITS BELOW: * TYPE * ** ENGLISH ** ** METRIC-KN ** ** METRIC-KG ** <table style="width: 100%; margin-top: 10px;"> <tbody> <tr> <td>FORCE</td> <td>KIP</td> <td>KN</td> <td>KG</td> </tr> <tr> <td>MOMENT</td> <td>KIP-FT</td> <td>KN-M</td> <td>KG-M</td> </tr> <tr> <td>DEFLECTION</td> <td>IN</td> <td>CM</td> <td>CM</td> </tr> <tr> <td>ROTATION</td> <td>RADIANS</td> <td>RADIANS</td> <td>RADIANS</td> </tr> </tbody> </table> ANY NUMBER OF POINTS CAN BE ENTERED BY REPEATING THIS DATA, LEAVING THE JOINTS BLANK ON SUBSEQUENT LINES.	FORCE	KIP	KN	KG	MOMENT	KIP-FT	KN-M	KG-M	DEFLECTION	IN	CM	CM	ROTATION	RADIANS	RADIANS	RADIANS
FORCE	KIP	KN	KG														
MOMENT	KIP-FT	KN-M	KG-M														
DEFLECTION	IN	CM	CM														
ROTATION	RADIANS	RADIANS	RADIANS														

LINE LABEL	BEGIN JOINT	END JOINT	DEGREE OF FREEDOM	1ST POINT		2ND POINT		3RD POINT		4TH POINT	
				FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION
NLSPJJ											
1-- 6	8-->11	12-->15	18--19	25<--31	32<--38	39<--45	46<--52	53<--59	60<--66	67<--73	74<--80

NONLINEAR SPRING INPUT

COLUMNSCOMMENTARY

GENERALTHIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY NONLINEAR SUPPORTS. ANY NUMBER OF NONLINEAR SUPPORTS CAN BE USED.

(8-11)ENTER THE JOINT NAME WHERE THE SUPPORT IS LOCATED.

(13-14)ENTER THE DEGREE OF FREEDOM FOR THIS CONSTRAINT FROM THE FOLLOWING:
'DX' - DEFLECTION IN GLOBAL X-DIRECTION
'DY' - DEFLECTION IN GLOBAL Y-DIRECTION
'DZ' - DEFLECTION IN GLOBAL Z-DIRECTION
'RX' - ROTATION ABOUT GLOBAL X-AXIS
'RY' - ROTATION ABOUT GLOBAL Y-AXIS
'RZ' - ROTATION ABOUT GLOBAL Z-AXIS

(15-78)ENTER THE FORCE-DEFLECTION OR MOMENT-ROTATION PAIRS OF VALUES IN ORDER OF INCREASING DEFLECTION/ROTATION USING THE UNITS BELOW:

* TYPE * ** ENGLISH ** ** METRIC-KN ** ** METRIC-KG **

FORCE KIP KN KG
MOMENT KIP-FT KN-M KG-M
DEFLECTION IN CM CM
ROTATION RADIANS RADIANS RADIANS

ANY NUMBER OF POINTS CAN BE ENTERED BY REPEATING THIS DATA, LEAVING THE JOINT NAME BLANK ON SUBSEQUENT LINES.

LINE LABEL	JOINT NAME	DEGREE OF FREEDOM	1ST POINT		2ND POINT		3RD POINT		4TH POINT	
			FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION
NLSPRG										
1-- 6	8-->11	13--14	15<--22	23<--30	31<--38	39<--46	47<--54	55<--62	63<--70	71<--78

COROTATIONAL NONLINEAR AXIAL SPRING INPUT

COLUMNS	COMMENTARY																
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SPECIFY A COROTATIONAL AXIAL NONLINEAR SPRING BETWEEN JOINTS. ANY NUMBER OF NONLINEAR AXIAL SPRINGS CAN BE USED.																
(8-11)	ENTER THE BEGIN JOINT OF THE SPRING.																
(12-15)	ENTER THE END JOINT OF THE SPRING.																
	NOTE THAT LOCAL COORDINATES ARE DEFINED USING THE UPDATED CO-ORDINATES. THIS IS IN CONTRAST TO TRADITIONAL NONLINEAR SPRINGS, WHICH HAVE LOCAL CO-ORDINATES DEFINED USING THE UNDEFORMED STATE OF THE STRUCTURE.																
(25-80)	ENTER THE FORCE-DEFLECTION OR MOMENT-ROTATION PAIRS OF VALUES IN ORDER OF INCREASING DEFLECTION/ROTATION USING THE UNITS BELOW:																
	* TYPE * ** ENGLISH ** ** METRIC-KN ** ** METRIC-KG ** <table style="width: 100%; margin-top: 5px;"> <tbody> <tr> <td>FORCE</td> <td>KIP</td> <td>KN</td> <td>KG</td> </tr> <tr> <td>MOMENT</td> <td>KIP-FT</td> <td>KN-M</td> <td>KG-M</td> </tr> <tr> <td>DEFLECTION</td> <td>IN</td> <td>CM</td> <td>CM</td> </tr> <tr> <td>ROTATION</td> <td>RADIANS</td> <td>RADIANS</td> <td>RADIANS</td> </tr> </tbody> </table>	FORCE	KIP	KN	KG	MOMENT	KIP-FT	KN-M	KG-M	DEFLECTION	IN	CM	CM	ROTATION	RADIANS	RADIANS	RADIANS
FORCE	KIP	KN	KG														
MOMENT	KIP-FT	KN-M	KG-M														
DEFLECTION	IN	CM	CM														
ROTATION	RADIANS	RADIANS	RADIANS														
	ANY NUMBER OF POINTS CAN BE ENTERED BY REPEATING THIS DATA, LEAVING THE JOINTS BLANK ON SUBSEQUENT LINES.																

LINE LABEL	BEGIN JOINT	END JOINT	1ST POINT		2ND POINT		3RD POINT		4TH POINT	
			FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION	FORCE OR MOMENT	DEFLECTION OR ROTATION
NLSPST										
1-- 6	8-->11	12-->15	25<--31	32<--38	39<--45	46<--52	53<--59	60<--66	67<--73	74<--80

ELASTIC PLATE GROUP SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO DESIGNATE PLATES AS ELASTIC.
(12-79)	ENTER THE PLATE GROUP IDENTIFIERS OF PLATES SELECTED AS ELASTIC. USE AS MANY OF THESE INPUT LINES AS DESIRED.

LINE LABEL	ELASTIC PLATE GROUP SELECTIONS													
	1ST GROUP	2ND GROUP	3RD GROUP	4TH GROUP	5TH GROUP	6TH GROUP	7TH GROUP	8TH GROUP	9TH GROUP	10TH GROUP	11TH GROUP	12TH GROUP	13TH GROUP	14TH GROUP
PGRELA														
1-- 6	12-->14	17-->19	22-->24	27-->29	32-->34	37-->39	42-->44	47-->49	52-->54	57-->59	62-->64	67-->69	72-->74	77-->79

ELASTIC PLATE SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO DESIGNATE PLATES AS ELASTIC.
(12-80)	ENTER THE PLATE IDENTIFIERS OF PLATES SELECTED AS ELASTIC. USE AS MANY OF THESE INPUT LINES AS DESIRED.

LINE LABEL	ELASTIC PLATE SELECTIONS													
	1ST PLATE	2ND PLATE	3RD PLATE	4TH PLATE	5TH PLATE	6TH PLATE	7TH PLATE	8TH PLATE	9TH PLATE	10TH PLATE	11TH PLATE	12TH PLATE	13TH PLATE	14TH PLATE
PLTELA														
1-- 6	12-->15	17-->20	22-->25	27-->30	32-->35	37-->40	42-->45	47-->50	52-->55	57-->60	62-->65	67-->70	72-->75	77-->80

PLATE SELECTION INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY COLLAPSE ANALYSIS RUN. IT IS USED TO SELECT PLATES FOR OUTPUT PRINT. IF OMITTED, THEN ALL PLATES ARE SELECTED BY DEFAULT.
(12-80)	ENTER THE PLATE IDENTIFIERS OF PLATES SELECTED FOR OUTPUT PRINT. USE AS MANY OF THESE INPUT LINES AS DESIRED.

LINE LABEL	PLATE SELECTIONS FOR OUTPUT REPORTS													
	1ST PLATE	2ND PLATE	3RD PLATE	4TH PLATE	5TH PLATE	6TH PLATE	7TH PLATE	8TH PLATE	9TH PLATE	10TH PLATE	11TH PLATE	12TH PLATE	13TH PLATE	14TH PLATE
PLTSEL														
1-- 6	12-->15	17-->20	22-->25	27-->30	32-->35	37-->40	42-->45	47-->50	52-->55	57-->60	62-->65	67-->70	72-->75	77-->80

ISO 19902 RESISTANCE FACTOR DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA ENABLES THE USER TO OVERRIDE THE RESISTANCE FACTORS SPECIFIED IN THE ISO 19902 STANDARD FOR STRENGTH OF TUBULAR JOINTS, NAMELY THE TUBULAR JOINT PARTIAL RESISTANCE FACTOR, THE YIELD STRENGH PARTIAL RESISANCE FACTOR AND THE EXTRA PARTIAL RESISTANCE FACTOR. FURTHERMORE, CONNECTION RESISTANCE FACTORS MAY BE SPECIFIED, THE DEFAULT VALUES FOR WHICH ARE 1.0.
(6-25)	ENTER THE CONNECTION RESISTANCE FACTORS FOR T AND Y TYPE JOINTS. IF ANY ITEM IS ENTERED AS ZERO OR LEFT BLANK, THEN ITS DEFAULT VALUE WILL BE USED.
(26-45)	ENTER THE CONNECTION RESISTANCE FACTORS FOR X TYPE JOINTS.
(46-65)	ENTER THE CONNECTION RESISTANCE FACTORS FOR K TYPE JOINTS.
(66-70)	ENTER THE PARTIAL RESISTANCE FACTOR FOR TUBULAR JOINTS.
(71-75)	ENTER THE PARTIAL RESISTANCE FACTOR FOR YIELD STRENGTH.
(76-80)	ENTER THE EXTRA PARTIAL RESISTANCE FACTOR.

LINE LABEL	T & Y JOINTS				X JOINTS				K JOINTS				TUBULAR JOINT RES. FAC	YIELD STRENGTH RES. FAC	EXTRA PARTIAL RES. FAC
	AXIAL TENS.	AXIAL COMP.	IN-PL BEND.	OUT-OF-PLANE BEND.	AXIAL TENS.	AXIAL COMP.	IN-PL BEND.	OUT-OF-PLANE BEND.	AXIAL TENS.	AXIAL COMP.	IN-PL BEND.	OUT-OF-PLANE BEND.			
RSFAC															
1-- 5	6<--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70	71<--75	76<--80
DEFAULT	1	1	1	1	1	1	1	1	1	1	1	1	1.05	1.05	1.17

LRFD RESISTANCE FACTOR DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA ENABLES THE USER TO OVERRIDE THE LRFD RESISTANCE FACTORS AS SPECIFIED IN THE API RP 2A-LRFD. THE DEFAULT VALUES ARE AS SPECIFIED IN THE COMMENTARY SECTION AS BETA FACTORS.
(6-25)	ENTER THE CONNECTION RESISTANCE FACTORS FOR T AND Y TYPE JOINTS. IF ANY ITEM IS ENTERED AS ZERO OR LEFT BLANK, THEN ITS DEFAULT VALUE WILL BE USED.
(26-45)	ENTER THE CONNECTION RESISTANCE FACTORS FOR X TYPE JOINTS.
(46-65)	ENTER THE CONNECTION RESISTANCE FACTORS FOR K TYPE JOINTS.
(66-70)	ENTER THE YIELD STRESS RESISTANCE FACTOR.

LINE LABEL	T & Y JOINTS				X JOINTS				K JOINTS				YIELD STRESS RESISTANCE FACTOR	LEAVE BLANK
	AXIAL TENS.	AXIAL COMP.	IN-PL BEND.	OUT-OF-PLANE BEND.	AXIAL TENS.	AXIAL COMP.	IN-PL BEND.	OUT-OF-PLANE BEND.	AXIAL TENS.	AXIAL COMP.	IN-PL BEND.	OUT-OF-PLANE BEND.		
RSFAC														
1-- 5	6<--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70	71--80
DEFAULT	2.11	2.57	2.81	2.61	2.11	2.57	2.81	2.61	2.51	2.51	2.81	2.61	1	

NORSOK RESISTANCE FACTOR DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA ENABLES THE USER TO OVERRIDE THE MATERIAL RESISTANCE FACTOR AS SPECIFIED IN THE NORSOK JOINT STRENGTH STANDARD. THE DEFAULT VALUE IS 1.15. FURTHERMORE, CONNECTION RESISTANCE FACTORS MAY BE SPECIFIED, THE DEFAULT VALUES FOR WHICH ARE 1.0.
(6-25)	ENTER THE CONNECTION RESISTANCE FACTORS FOR T AND Y TYPE JOINTS. IF ANY ITEM IS ENTERED AS ZERO OR LEFT BLANK, THEN ITS DEFAULT VALUE WILL BE USED.
(26-45)	ENTER THE CONNECTION RESISTANCE FACTORS FOR X TYPE JOINTS.
(46-65)	ENTER THE CONNECTION RESISTANCE FACTORS FOR K TYPE JOINTS.
(66-70)	ENTER THE MATERIAL RESISTANCE FACTOR.

[illegible]

BRACE RESISTANCE FACTOR OVERRIDE

COLUMNS	COMMENTARY
GENERAL	THIS DATA ENABLES THE USER TO OVERRIDE THE LRFD RESISTANCE FACTORS AS SPECIFIED IN THE API RP 2A-LRFD FOR AN INDIVIDUAL BRACE. OVERRIDES SPECIFIED ON THE 'RSFACO' LINE REPLACE ANY RESISTANCE FACTOR OVERRIDES SPECIFIED ON THE 'RSFAC' LINE FOR THE BRACE END SPECIFIED. IF ANY FACTOR IS ENTERED AS ZERO OR LEFT BLANK, THEN THE VALUE ENTERED ON THE 'RSFAC' LINE IS USED.
(1- 6)	ENTER 'RSFACO'
(8-11)	ENTER THE BEGIN JOINT OF THE BRACE TO WHICH THE OVERRIDES APPLY.
(12-15)	ENTER THE END JOINT OF THE BRACE TO WHICH THESE OVERRIDES APPLY.
(16-19)	ENTER THE JOINT DESIGNATING THE BRACE CONNECTION TO WHICH THESE OVERRIDES APPLY. ENTER THE BEGIN JOINT TO CONSIDER THE CONNECTION AT THE START OF THE BRACE OR THE END JOINT TO CONSIDER THE CONNECTION AT THE END OF THE BRACE.
(21-25)	ENTER THE AXIAL TENSION RESISTANCE FACTOR.
(26-30)	ENTER THE AXIAL COMPRESSION RESISTANCE FACTOR.
(31-35)	ENTER THE IN-PLANE BENDING RESISTANCE FACTOR.
(36-40)	ENTER THE OUT-OF-PLANE BENDING RESISTANCE FACTOR.
(41-45)	ENTER THE YIELD STRENGTH RESISTANCE FACTOR.
(47)	OPTIONALLY SPECIFY THE CONNECTION TYPE TO WHICH THE OVERRIDE FACTORS APPLY AS FOLLOWS: 'X' - X OR CROSS CONNECTION 'Y' - T & Y CONNECTION 'K' - K BRACE CONNECTION

LINE LABEL	BRACE			RESISTANCE FACTOR OVERRIDES					CONNECTION TYPE	LEAVE BLANK
	BEGIN JOINT	END JOINT	CONNECTION	AXIAL TENS.	AXIAL COMP.	IN-PLANE BEND.	OUT-OF-PLANE BEND.	YIELD STRESS		
RSFACO										
1-- 6	8-->11	12-->15	16-->19	21--25	26--30	31--35	36--40	41--45	47	48--80

SHIP INDENTATION CURVE INPUT

COLUMNS	COMMENTARY
GENERAL	THIS OPTIONAL LINE SET IS USED TO DEFINE A SHIP INDENTATION CURVE. THE SHIP INDENTATION CURVE IS REFERENCED BY THE 'IMPACT' LINE. FOR SHIP INDENTATION CURVES WITH MANY FORCE/INDENTATION PAIRS, THIS LINE IS REPEATED WITHOUT THE INDENTATION CURVE NAME UNTIL THE CURVE DEFINITION IS COMPLETE.
(8-11)	ENTER THE INDENTATION CURVE IDENTIFIER ON THE FIRST LINE OF A 'SHPIND' LINE SET.
(13-18)	ENTER THE SHIP FORCE.
(20-24)	ENTER THE SHIP INDENTATION. SUBSEQUENT FORCE/INDENTATION PAIRS ARE ENTERED SIMILARLY. A BLANK FORCE/INDENTATION PAIR IS SKIPPED.

LINE LABEL	INDENTATION CURVE NAME	FORCE	SHIP INDENTATION	FORCE	SHIP INDENTATION	FORCE	SHIP INDENTATION	FORCE	SHIP INDENTATION	FORCE	SHIP INDENTATION
SHPIND											
1-- 6	8-->11	13<--18	20<--24	26<--31	33<--37	39<--44	46<--50	52<--57	59<--63	65<--70	72<--77
DEFAULT											
ENGLISH		KIP	FT	KIP	FT	KIP	FT	KIP	FT	KIP	FT
METRIC (KN)		MN	M	MN	M	MN	M	MN	M	MN	M
METRIC (KG)		TONNE	M	TONNE	M	TONNE	M	TONNE	M	TONNE	M

YIELD STRESS FACTOR

COLUMNS	COMMENTARY
GENERAL	THIS DATA ENABLES THE USER TO MODIFY THE YIELD STRESS OF ALL ENTITIES BY SPECIFYING A YIELD STRESS FACTOR. WITH THIS LINE, THE MODIFIED YIELD STRESS USED BY COLLAPSE IS THE YIELD STRESS SPECIFIED IN THE SACS IV MODEL MULTIPLIED BY THIS FACTOR. THIS IS PARTICULARLY USEFUL IN PROVIDING THE MEANS OF SUPPLYING A DYNAMIC INCREASE FACTOR IN BLAST ANALYSIS.
(8-13)	ENTER THE YIELD STRESS FACTOR.

LINE LABEL	YIELD STRESS FACTOR	LEAVE BLANK
YSFACT		
1-- 6	8<--13	14-----80
DEFAULT	1	

YIELD STRESS MEMBER GROUP OVERRIDE LINE

COLUMNS

COMMENTARY

GENERAL THE 'YSMGOV' LINE IS USED TO OVERRIDE THE SACS IV MODEL YIELD STRESS FOR SPECIFIED MEMBER GROUPS. UP TO 15 MEMBER GROUPS MAY BE OVERRIDDEN ON ONE LINE.

(8-13) ENTER THE REPLACEMENT YIELD STRESS. ALL MEMBER GROUPS SPECIFIED WILL HAVE THEIR YIELD STRESS VALUES IN COLLAPSE SET EQUAL TO THIS VALUE.

(15-73) ENTER MEMBER GROUPS.

[illegible]

YIELD STRESS MODIFICATION LINE

COLUMNS	COMMENTARY
GENERAL	THE 'YSUMOD' LINE IS USED TO REPLACE A SINGLE SACS IV MODEL YIELD STRESS VALUE WITH A NEW VALUE. THIS LINE IS OVERRIDDEN BY THE 'YSMGV' LINE.
(8-13)	ENTER THE FIRST YIELD STRESS VALUE TO BE REPLACED FROM THE SACS IV MODEL.
(14-19)	ENTER THE COLLAPSE YIELD STRESS VALUE TO REPLACE THE SACS IV MODEL VALUE FROM COLUMNS 8-13.
(20-25)	ENTER THE SECOND YIELD STRESS VALUE TO BE REPLACED FROM THE SACS IV MODEL.
(26-31)	ENTER THE COLLAPSE YIELD STRESS VALUE TO REPLACE THE SACS IV MODEL VALUE FROM COLUMNS 20-25.
(32-37)	ENTER THE THIRD YIELD STRESS VALUE TO BE REPLACED FROM THE SACS IV MODEL.
(38-43)	ENTER THE COLLAPSE YIELD STRESS VALUE TO REPLACE THE SACS IV MODEL VALUE FROM COLUMNS 32-37.
(44-49)	ENTER THE FOURTH YIELD STRESS VALUE TO BE REPLACED FROM THE SACS IV MODEL.
(50-55)	ENTER THE COLLAPSE YIELD STRESS VALUE TO REPLACE THE SACS IV MODEL VALUE FROM COLUMNS 44-49.
(56-61)	ENTER THE FIFTH YIELD STRESS VALUE TO BE REPLACED FROM THE SACS IV MODEL.
(62-67)	ENTER THE COLLAPSE YIELD STRESS VALUE TO REPLACE THE SACS IV MODEL VALUE FROM COLUMNS 56-61.

[illegible]

YIELD STRESS UNIVERSAL OVERRIDE LINE

COLUMNS	COMMENTARY
GENERAL	THE 'YSUOVR' LINE IS USED TO REPLACE ALL INSTANCES OF A SACS IV MODEL YIELD STRESS WITH THE SAME VALUE. THIS LINE IS OVERRIDDEN BY THE 'YSMGOV' AND 'YSUMOD' LINES.
(8-13)	ENTER THE YIELD STRESS VALUE.

LINE LABEL	REPLACEMENT YIELD STRESS	LEAVE BLANK
YSUOVR		
1-- 6	8<--13	14-----80
DEFAULT		
ENGLISH	KSI	
METRIC (KN)	KN/SQ.CM	
METRIC (KG)	KG/SQ.CM	